Predicting Physical Links in Networks

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Physical links are fundamental to the structure, function, and ultimately the understanding of many networked systems. Erroneously identified, such direct cause-effect relations undermine the understanding of the system's dynamics and can impede effective intervention. To address the long-standing challenge of distinguishing direct from indirect links in observational data from complex dynamical networks, we introduce a general inference framework tailored to networked dynamical systems. Our approach combines cross mapping with a novel, physics-informed technique for incremental variable construction that leverages temporal delay effects to identify relevant variables and eliminate indirect links. We validate our method across a diverse set of noisy benchmark and real-world systems, including brain networks, gene regulation, synthetic gene circuits, and biosphere-atmosphere interactions. The method achieves particularly strong gains on mesoscale networks ($n \approx 5$–20), consistently outperforming existing approaches under diverse noise conditions. Although designed for small- to medium-sized systems, it continues to outperform competing methods even in larger networks, demonstrating both robustness and broad applicability.

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I. INTRODUCTION

The temporal evolution of many complex systems can be described as a networked dynamical system, where nodes influence one another through nonlinear interactions. Examples include neural circuits, gene regulatory networks, and ecosystems [1–3]. A central challenge is to infer the *direct* physical links from time-series data, despite pervasive indirect interactions and finite-sample effects [4,5].

Many widely used approaches rely on conditional independence (CI) logic to separate direct from indirect dependencies. However, CI-based methods require strong assumptions (e.g., model class, faithfulness, or noise structure), and recent theoretical work shows that no CI test is universally valid [6]. As a consequence, CI-based causal discovery can be brittle: its performance depends sensitively on the underlying dynamical regime and on whether its assumptions are met.

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An alternative line of work exploits state-space reconstruction. Under mild assumptions, Takens' theorem [7,8] implies that the dynamics of an interacting system can be reconstructed from a single observed variable using time-delay embedding. In practice, embeddings with dimensions close to the attractor dimension often recover the relevant geometry even in the presence of modest noise [8–10]. This insight motivates *cross-mapping* methods, which infer interactions by testing whether the historical states of one variable can be reconstructed from another: a cause leaves an imprint in the effect's reconstructed state space (a shadow manifold). Latent cross mapping [11] further extends this idea to partially observed systems.

Despite these advances, accurately identifying *direct* physical links in nonlinear networks remains difficult. First, interaction transitivity can yield spurious links: indirect pathways can produce strong apparent dependence even when no direct link exists [9,10]. Second, computational cost grows quickly with system size and embedding choices, limiting scalability [12–14]. Third, in nonlinear settings variables may be non-separable, meaning their reconstructed state spaces overlap and their individual contributions cannot be cleanly disentangled [15]. Recent methods attempt to mitigate these issues, for example, by combining the Peter-Clark algorithm with momentary CI test (PCMCI) [16] or by using multiview distance-regularized S maps for high-dimensional reconstruction [17]. Nonetheless, the combination of transitivity, nonseparability, and moderate dimension remains a key obstacle for reliable link inference.

These challenges are especially prominent in mesoscale networks ($n \approx 5$–20), which arise when a limited number of interacting components generate rich, time-dependent dynamics. Examples include directed coupling among hippocampal CA1 neurons [18], yeast oxidative-stress gene-regulatory circuits [19], multichannel ICU recordings where respiration- and heart-driven signals must be separated [20], feedback loops in smart buildings [21], and tightly coupled microgrids where delay-based protocols can outperform PCMCI [22]. These systems share two key features: a moderate number of variables and nontrivial temporal delays.

To infer direct physical links under exactly these conditions, we propose incremental variable construction-based cross mapping (InVaXMap). The method integrates cross mapping with a physics-informed strategy for incremental variable construction. The core consideration is simple: direct effects typically manifest on shorter timescales than indirect cascades. Direct pathways therefore produce stronger cross mapping at short lags, whereas indirect pathways accumulate additional delays and dispersion, weakening correlations at longer lags. Guided by this intuition, InVaXMap ranks candidate interactions by their optimal delay and constructs a conditioning set from shortest to longest delays, filtering indirect links without overconditioning (Fig. 1).

We evaluate InVaXMap on synthetic benchmarks that mirror mesoscale complexity, including discrete Lotka-Volterra competition models, Ricker population dynamics, and coupled Lorenz attractors, as well as on empirical datasets spanning brain activity under alcohol exposure [23], DREAM4 gene expression time series [24], FLUXNET2015 biosphere-atmosphere fluxes [25], a well-characterized synthetic gene circuit [26] and concentrations of air pollutants [27]. Across these systems, InVaXMap improves link inference accuracy over existing approaches, and its advantage persists as system size increases.

II. METHODS

In this section, we present incremental variable construction-based cross mapping, our approach for physical link inference in dynamical networked systems. Our method employs extended convergent cross mapping (ECCM) and *incremental variable construction* to isolate direct physical links while mitigating redundant influences. The core idea of InVaXMap is to iteratively construct a conditioning set of variables based on their estimated time delays and partial correlation indices. By incrementally selecting variables in a causally coherent order, our method efficiently distinguishes between *direct* causal pathways and spurious correlations (Fig. 1).

A. Rationale for incremental variable construction

Consider a delayed autonomous dynamical system with N variables, $X = \{X_1, \ldots, X_N\}$, plus some small noise σ_t,

$$\dot{X}(t) = F[X(t-\tau_1), X(t-\tau_2), \ldots, X(t-\tau_k)] + \sigma_t, \qquad (1)$$

where $\tau_k > \cdots > \tau_2 > \tau_1 > 0$ are k time delays. We assume that the steady states of the system are constrained on a low-dimensional attractive manifold. After integration, time truncation, and numerical discretization, for some discrete Δ and some small noise ϵ_t, Eq. (1) transforms into [28]

$$X(t) = \hat{F}[X(t-1), X(t-2), \ldots, X(t-\Delta)] + \epsilon_t. \quad (2)$$

In this system, we assume that all variables are observed, and that causal effects between variables, as well as the time delay for causal effects, remain unchanged over time.

Causal discovery theory on dynamic systems [15] states that whenever a variable x influences another variable y, information about x is embedded in y and can be recovered. This follows from the mapping between the original manifold and the shadow manifold created from lags of y. Generalized Takens' theorems [8] have proven that maps created from lags of multiple observation functions are generically embeddings. Consequently, information about x is encoded in the variables it influences and can be reconstructed using these multivariate time series with corresponding lags.

Thus, a set of variables with specific time delays that satisfy this condition can be selected and used as the conditioning set to detect direct physical links.

Each variable influenced by x filters information about x differently. To enhance the credibility of interpretations, we must incorporate more variables that might explain an indirect link, but this can lead to reduced power in detecting direct links due to higher dimensionality and information redundancy [16]. Therefore, we aim to condition only on the few mediator variables that actually explain an indirect link. Since mediator variables generally exhibit smaller time delays than irrelevant variables when they lie on the same path, combining these signals in an incremental manner allows us to construct a conditioning set that is very likely to include the mediator variables while filtering out irrelevant ones. In other words, we can capture a more complete representation with minimal redundancy. This ordering ensures a balance between completeness and efficiency in reconstructing the physical influences.

Moreover, we note that the conditioning set for detecting direct physical links is nonunique and may

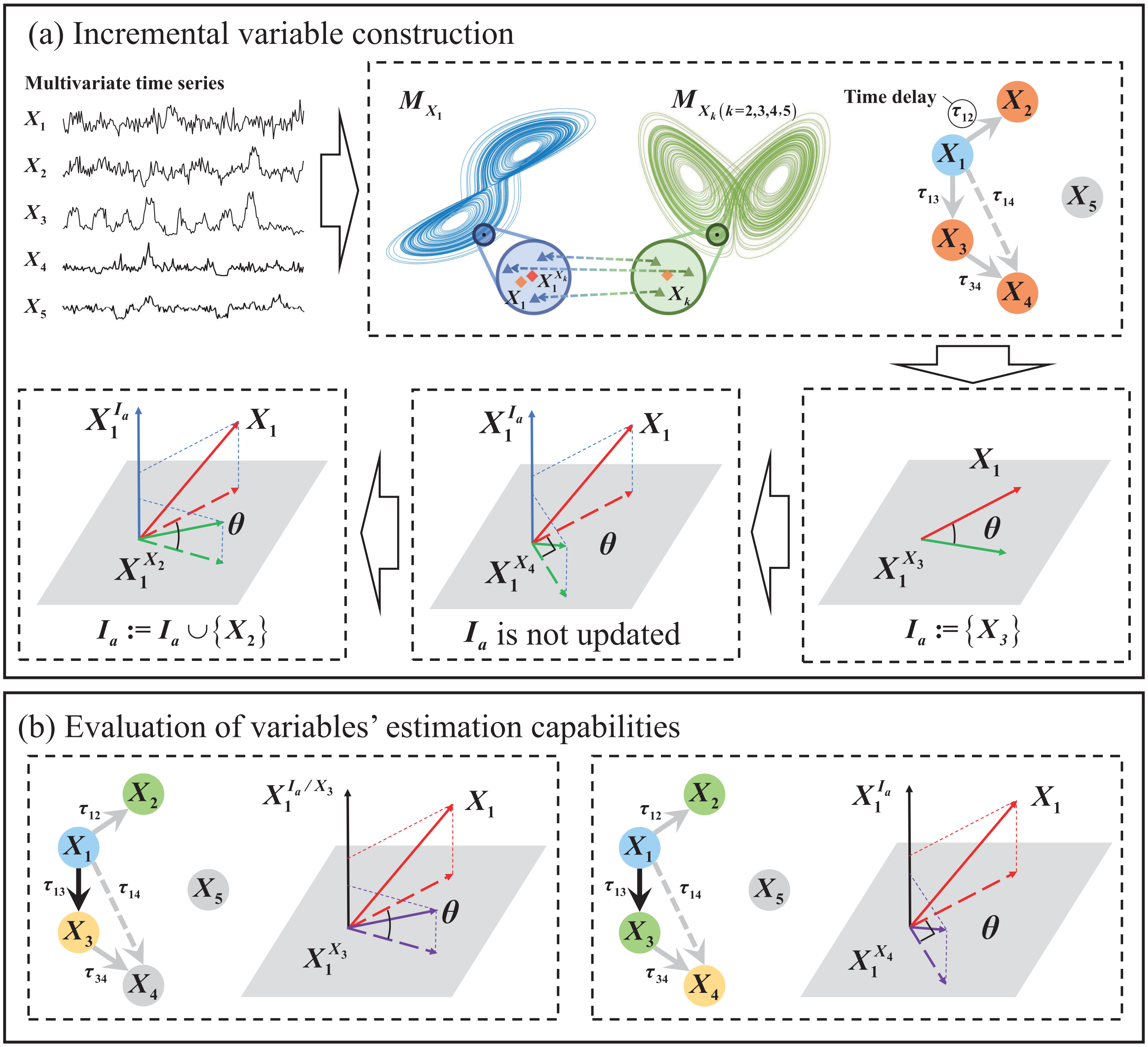

FIG. 1. Framework of InVaXMap. (a) Incremental variable construction: The objective is to construct a set of variables with specific time delays to serve as the conditioning set. The extended convergent cross mapping method is used to identify physically relevant variables, which are ranked according to their associated time delays ($\tau_{13} < \tau_{14} < \tau_{12}$). The correlation index $\cos(\theta) = \rho_{X_1 \to X_k}$ or the partial correlation index $\cos(\theta) = \rho^{I_a}_{X_1 \to X_k}$ measures the alignment between the target variable X_1 and its estimation from another variable (e.g., $X_1^{X_k}$). The set I_a is incrementally updated: initially, $I_a \coloneqq \{X_3\}$ since $\rho_{X_1 \to X_3} > \epsilon$; however, $\rho^{I_a}_{X_1 \to X_4} \le \epsilon$, meaning X_4 is not included. The final update yields $I_a \coloneqq \{X_3, X_2\}$ as $\rho^{I_a}_{X_1 \to X_2} > \epsilon$. After this process, I_a contains all mediator variables, which transmit physical influence from X_1 to other nodes. The reconstructed shadow manifolds of nodes X_k are denoted as M_{X_k}, representing their phase-space trajectories reconstructed from time-delay embeddings. (b) Evaluating estimation capability: The estimation capability of each variable is assessed using a partial correlation index, quantifying how well X_1 can be estimated from another variable under a conditioned set. If $X_3 \in I_a$, its estimation capability is measured as the partial correlation between X_1 and $X_1^{X_3}$, conditioned on $X_1^{I_a/X_3}$. For variables outside I_a (e.g., X_4 and X_5), their estimation capability is computed similarly but conditioned on $X_1^{I_a}$. The angle θ represents the difference in alignment between X_1 and its estimated reconstruction from another variable, indicating the reliability of inference. Graph representation: The black node (X_1) is the target variable. Orange nodes indicate variables initially identified as physically relevant via ECCM. Green nodes represent the selected variables in I_a, while yellow nodes correspond to variables under evaluation. Black links indicate direct physical links detected by InVaXMap, gray links denote potential causal relationships, solid lines represent direct physical effects, and dashed lines indicate indirect physical influences.

ALGORITHM 1. Pseudocode of InVaXMap

Input: Time series $X=[X_1,X_2,\ldots,X_n]^T, X_i=[x_{11},\ldots,x_{1L}]\in R^L$, maximum time delay $\tau_{\max}$, embedding dimension E and threshold ϵ.
Output: Link matrix C_F.

Select target variable X_i, then set $I_{\text{inactive}}=\varnothing$ and $I_a=\varnothing$
for all $j=1,2,\ldots,n$ and $j\neq i$ **do**
  for $\tau=0,1,\ldots,\tau_{\max}$ **do**
    Carry out cross mapping technique to estimate $X_i=[x_{i1},\ldots,x_{i,L-\tau_{\max}}]$ using
    $X_{j,\tau}=[x_{j,1+\tau},\ldots,x_{j,L-\tau_{\max}+\tau}]$, then get $X_{i,\tau}^{X_j}$ and effect $\rho_{ij,\tau}$ $X_{i,\tau}^{X_j},\rho_{ij,\tau}=\text{CMT}(X_i,X_{j,\tau},E)$
  end for
  Let $\rho_{ij}=\max\{\rho_{ij,\tau}\}$, $\tau_{ij}=\arg_\tau\{\rho_{ij,\tau}=\rho_{ij}\}$ and $X_i^{X_j}=X_{i,\tau_{ij}}^{X_j}$
  if $\rho_{ij}>\epsilon$ **then**
    Put X_j in I_{inactive}.
  end if
end for
while $I_{\text{inactive}}\neq\varnothing$ **do**
  Drop out X_j in I_{inactive} which satisfied $X_j=\arg\min_j\{\tau_{ij},X_j\in I_{\text{inactive}}\}$
  if $I_a=\varnothing$ **then**
    $I_a=I_a\cup\{X_j\}$
  else
    Calculate partial correlation index $\rho_{X_1\to X_j}^{I_a}$
    if $\rho>\epsilon$ **then**
      Put X_j in I_a.
    end if
  end if
end while
for all $j=1,2,\ldots,n$ and $j\neq i$ **do**
  Calculate partial correlation index $C_F(i,j)=\rho_{X_1\to X_j}^{I_a/X_j}$ with chosen measurement.
end for
Change target variable X_i untill each variable was selected.

consist of different variables with corresponding lags. Incremental variable construction, as employed in our method, represents one possible way, yet a very efficient one, to identify such a set. Other criteria for selecting conditioning sets could be explored to address specific research objectives (Appendix E 7).

B. InVaXMap framework

The framework is summarized in the diagram below (pseudocode in Algorithm 1).

Incremental variable construction-based cross mapping

(1) *Reconstruct shadow manifolds*
  (i) Use *delay embedding* to reconstruct the phase space for each variable.
(2) *Apply ECCM to identify descendants*
  (i) Compute the cross-map correlation between X_i and X_j over various time lags l.
  (ii) Maximize the correlation over l to determine the strongest physical link.
(3) *Incremental variable construction*
  (i) Identify a set of *informative variables*.
  (ii) Order them based on their estimated time delays.
(4) *Compute partial correlations*
  (i) Use *partial correlation* to remove indirect influences.
  (ii) Take the absolute value of the partial correlation index to assess the magnitude of influence.
(5) *Thresholding for links*
  (i) A physical link is confirmed if the correlation exceeds a threshold ϵ.

C. Extended convergent cross mapping

Here, we review ECCM, which we employ in our method. Consider a dynamic system of N variables $\{X_1, X_2,\ldots,X_N\}$. Denote the time series of length L generated by this system as $X_i=\{X_i(t)\}_{t=1}^L, i=1,2,\ldots,N$.

The corresponding N shadow manifolds in the delay-coordinate space, one from each X_i, can be reconstructed based on Takens' embedding theorem [7,8],

$$M_{X_k}=\{\bar{X}_k(t)\}_{t=1+(E-1)\tau}^L. \tag{3}$$

Here, E is the embedding dimension, satisfying $E>2d$, where d is the box-counting dimension of the original system, and τ is the time lag.

ALGORITHM 2. Pseudocode of cross mapping technique

Input: Two series X, Y and embedding dimension E.
Output: Estimation series $\hat{X}$ and effect ρ.
1: **function** CMT(X, Y, E)
2: Build $M_Y = \{\bar{y}(t)\}_{t=1+(E-1)\tau}^{L}$, where τ is time lag usually setting $\tau = 1$

$$\bar{y}(t) = [y(t), y(t-\tau), \cdots y(t-(E-1)\tau)]^T$$

3: Find $E+1$ nearest neighbors $\bar{y}(t_1), \bar{y}(t_2), \ldots, \bar{y}(t_{E+1})$ of each $\bar{y}(t)$
4: Calculate

$$\hat{x}(t) = \sum_{i=1}^{E+1} w_i x(t_i), \qquad t = 1+(E-1)\tau, \ldots, L$$

where

$$w_i = u_i / \sum u_j, \qquad u_j = \exp\{-d[\bar{y}(t), \bar{y}(t_j)]/d[\bar{y}(t), \bar{y}(t_1)]\}$$

$d(x, y)$ measures the Euclidean distance between x and y.
5: **return** estimation series $\hat{X} = \{\hat{x}(t)\}_{t=1+(E-1)\tau}^{L}$, effect $\rho = \mathrm{corr}(X, \hat{X})$ where $\mathrm{corr}(\cdot,\cdot)$ denotes Pearson correlation coefficient.
6: **end function**

The delay-coordinate vector is given by

$$\bar{X}_i(t) = [X_i(t), X_i(t-\tau), \ldots, X_i(t-(E-1)\tau)]^T. \quad (4)$$

For each point $\bar{X}_j(t)$, one can find the $E+1$ nearest neighbors on M_{X_j}, denoted as $\bar{X}_j(t_1), \bar{X}_j(t_2), \ldots, \bar{X}_j(t_{E+1})$. The estimation of $X_i(t)$, denoted as $X_i(t)^{X_j(t)}$, is obtained via a locally weighted mean of $X_i(t_k)$,

$$X_i(t)^{X_j(t)} = \sum_{k=1}^{E+1} w_k X_i(t_k), \quad (5)$$

where $w_k = u_k / \sum_{k'=1}^{E+1} u_{k'}$. $u_{k'} = \exp(-d[\bar{X}_j(t), \bar{X}_j(t_{k'})]/d[\bar{X}_j(t), \bar{X}_j(t_1)])$ and $d(x, y)$ is the Euclidean distance between two vectors x and y.

To identify the strongest causal effect of X_i on X_j across different time delays, we determine the maximum correlation coefficient between time series $X_i(t)$ and $X_i(t)^{X_j(t+l)}$, computed over a candidate range of delays $[0, l_{\max}]$:

$$\rho_{X_i \to X_j} = \max_l \left| \mathrm{corr}\left(X_i(t), X_i(t)^{X_j(t+l)}\right) \right|. \quad (6)$$

Since causal effects manifest after their causes with a temporal delay, the effect of X_i on X_j may appear strongest at a specific lag l_{ij}, which is chosen according to Eq. (6) as $l_{ij} = \arg\max_l \rho_{X_i \to X_j}$.

A significant interaction from X_i to X_j is established if $\rho_{X_i \to X_j}$ exceeds an empirical threshold ϵ.

D. Incremental variable construction-based cross mapping

Incremental variable construction and link inference through estimation capabilities are the two key principal steps.

Step 1: Incremental variable construction: To construct a set of causally informative variables, we first apply ECCM [29] to identify and eliminate irrelevant variables for a given target variable. The set of remaining variables, denoted as I_{inactive}, contains all informative descendants of the target variable. ECCM also estimates the corresponding time delays, which guide variable selection, as illustrated in Fig. 1(a).

The estimation capability of a variable Y with respect to the target variable X_1 is measured by the correlation index $\rho_{X_1 \to Y}$, computed using ECCM. Some variables in I_{inactive} may appear highly informative due to indirect information flow through mediator variables. To filter out such indirect influences, we perform iterative cross-map-based testing, systematically assessing a variable's estimation capability while controlling for potential mediators [9,14].

Since mediator variables tend to have smaller time delays than irrelevant variables on the same causal path, we rank variables in I_{inactive} in ascending order of time delay. The set of confirmed mediator variables, denoted as I_a, is initialized as empty. At each iteration, we select the variable X_j with the smallest time delay:

$$j = \arg\min_k \{\tau_{1k} | X_k \in I_{\text{inactive}}\}. \quad (7)$$

We compute the partial correlation index $\rho^{I_a}_{X_1\to X_j}$, using both linear (ParCorr) and nonlinear (GPDC) measurements, to assess the relationship between X_1 and X_j while conditioning on I_a. If $\rho^{I_a}_{X_1\to X_j}$ exceeds a threshold ϵ, then X_j provides information about X_1 beyond what is already in I_a. We then add X_j to I_a. We repeat until no remaining variable passes the test; equivalently, all unique information has been captured and I_{inactive} is empty.

Step 2: Link inference via estimation capability: Once I_a is constructed, we infer direct causal relationships by evaluating the estimation capability of each variable with respect to the target variable.

For each $X_j \in I_a$, we compute the partial correlation index:

$$\rho^{I_a\setminus X_j}_{X_1\longrightarrow X_j}, \tag{8}$$

which conditions on all other variables in I_a. If $X_j \notin I_a$, we compute

$$\rho^{I_a}_{X_1\to X_j} \tag{9}$$

to measure causality from X_1 to X_j.

Since high-dimensional systems often include irrelevant variables, conditioning on the full set of observed variables can reduce the power of link detection [16]. To mitigate this issue, we restrict conditioning to only the identified relevant variables, ensuring robustness in high-dimensional settings.

E. Computational complexity

Let N be the number of variables (network dimension, number of nodes), T the length of the observed time series, and L the maximum time lag considered ($\tau_{\max}$).

In the worst-case of a fully connected network, our method performs $N(N-1)L$ cross-mapping operations in the first stage (incremental variable construction), each requiring $O(T)$ work to compute embedding distances and correlations, yielding the cost of $O(N^2TL)$. Then, there are $N(N-2)$ partial-correlation tests in the first stage and $N(N-1)$ tests in the second stage (link inference via estimation capability), each with at least $O(T)$ cost, contributing an additional $O(N^2T)$ term. Hence, for fixed embedding and threshold parameters, the total *average-case* complexity of our algorithm is expected to be no better than

$$O(N^2TL + N^2T).$$

By comparison, ECCM requires only the cross-mapping stage with $O(N^2TL)$, so our method incurs no better than an extra $O(N^2T)$ overhead but achieves substantially lower false discovery rates in moderate-sized systems ($N \lesssim 100$).

TABLE I. Computation times (in seconds) of three embedding-based methods, ECCM, PCM, and InVaXMap-ParCorr, on systems with size $N = 10$, 50, and 100 (PCM is unable to produce results on systems with size 100).

	$N = 10$	$N = 50$	$N = 100$
ECCM	19.75	539.74	2176.44
PCM	128.77	9814.67	-
InVaXMap-ParCorr	21.88	685.69	2382.04

Clearly, however, the computational complexity of our method and others depends on the data, the network structure, and the empirical threshold.

In particular, PCMCI's complexity depends on the maximum conditioning-set size and can exceed $O(N^3TL^2 + N^2TL)$ in dense regimes, but is comparable or lower when the effective neighborhood is small [16]. Thus, although slower than ECCM in large N limits, our approach offers a favorable accuracy-efficiency trade-off for moderately large networks. We also recorded computation times for three embedding-based methods—ECCM, PCM, and InVaXMap-ParCorr—on the same time series ($L = 100$) using identical settings (embedding dimension, time lag, and maximum delay). For each system size, we ran 20 independent trials and report the average runtime (Table I). InVaXMap-ParCorr is consistently slower than ECCM, but substantially faster than PCM.

III. RESULTS

A. Detecting physical links in benchmark systems

To validate the effectiveness of our method, we carried out experiments for the benchmark system of multispecies discrete Lotka-Volterra competition model,

$$x_i(t+1) = x_i(t)\Big(r_i - \sum a_{ij}x_j(t-\tau_{ij})\Big) + \eta_i(t), \tag{10}$$

where $x_i(t)$ is the abundance of species i at time t, r_i is the growth rate of x_i, a_{ij} is the interaction strength, τ_{ij} is the time delay and $\eta_i(t)$ is process noise. We applied InVaXMap to a three-species system in which there are links from X_1 to X_2, X_2 to X_3, and X_3 to X_1, respectively. The partial correlation index for the three spurious links are insignificantly small, while significantly large for the other identified links as shown in Appendix E 2, Fig. 12. Thus, our method robustly removes spurious links induced by transitivity. Additionally, we applied our method to more complex interacting modes of three-species systems, known as the network motifs: cascading, fan-out and fan-in modes. InVaXMap robustly identifies physical links across a wide range of threshold values, indicated by the separation of the symbols representing pairs of species with physical links from those representing pairs of species without physical links (Appendix E 2, Fig. 13).

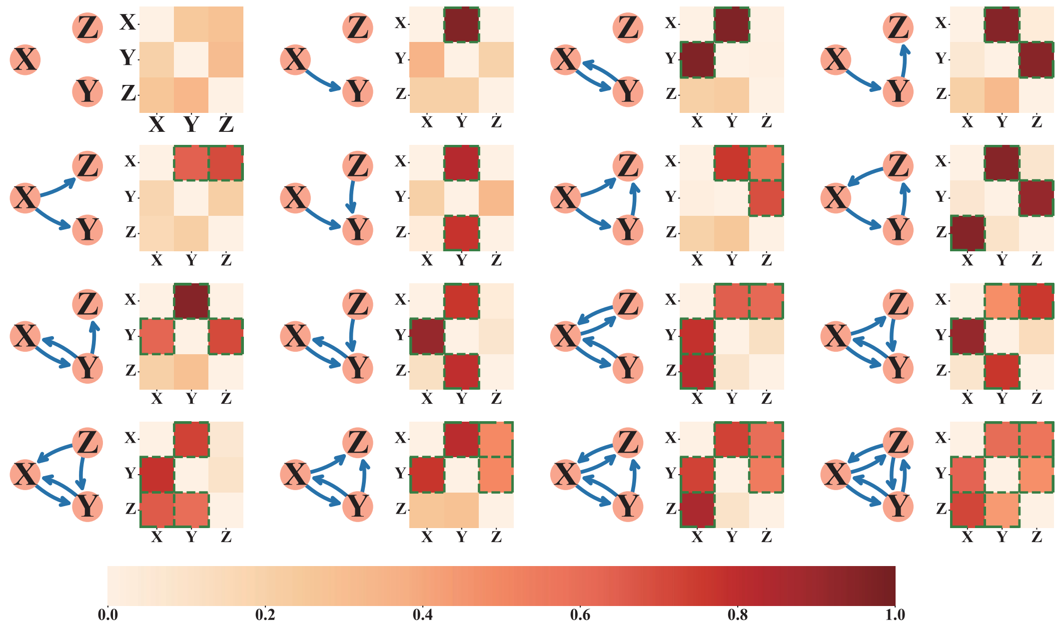

FIG. 2. Physical link detection for all interaction modes of three-species systems. For each interaction mode the actual true structure is on the left panel while the results obtained from InVaXMap is on the right panel. Here, InVaXMap is able to identify physical links correctly and completely. The areas encompassed by green dashed lines are the identified physical relationships.

Furthermore, we applied our method to all possible interacting modes of three-species systems. Our results show that InVaXMap accurately identifies physical structures for all these interacting modes of three-species systems, as shown in Fig. 2. The threshold value at which the accuracy of InVaXMap achieves over 90% ranges from 0.27 to 0.59, which accounts for 32% of the whole threshold interval, indicating that the performance of InVaXMap is robust against a wide range of threshold values (Fig. 11).

To further validate the effectiveness of InVaXMap, we studied model networks in three specific structural modes: cycle mode, random mode, and a cycle structure combined with random interactions (structural stochastic mode), as illustrated in Figs. 3(a), 3(c), and 3(e), respectively. We conducted systematic comparisons between InVaXMap and other state-of-the-art causality detection methods based on state space reconstruction, specifically ECCM and PCM methods. The results are as follows: As shown in Fig. 3(b), ECCM performs the worst in cycle mode, with an average AUC below 0.6 across various system sizes. PCM performs well for small system sizes, but its AUC drops significantly when the system size exceeds 60, and it is unable to produce results when the system size exceeds 70. This drop in performance is due to PCM conditioning on too many irrelevant variables in large systems, leading to diminished detection power. In contrast, InVaXMap consistently outperforms both ECCM and PCM across all system sizes. For both the random mode and the structural stochastic mode, the performance of ECCM and PCM deteriorates as the system size increases, with PCM failing to produce results once the system size surpasses 80. However, InVaXMap maintains strong performance throughout the entire range of system sizes, achieving the highest average AUC, as shown in Figs. 3(d) and 3(f).

We also evaluated other classical and state-of-the-art causality detection methods that are not based on state space reconstruction (Appendix E 3, Fig. 14). Across all comparisons, InVaXMap demonstrates the best performance for the cycle mode, random mode, and structural stochastic mode (Appendix E 3 Tables II, III, and IV). Overall, our results highlight the effectiveness of the InVaXMap method in detecting physical links within high-dimensional nonlinear systems, particularly in challenging scenarios where the system variables are nonseparable and weakly coupled.

To evaluate the robustness of InVaXMap across varying time series lengths and different levels of Gaussian observation noise, we conducted tests under multiple noise strengths. We applied InVaXMap to the benchmark systems with different system sizes. First, we varied the length of the time series while keeping the noise strength fixed at 0.002. As shown in Figs. 4(a), 4(b), and 4(c), the link detection power of InVaXMap remains consistently high across different system sizes and improves as the length of the time series increases. Next, we fixed the time series length at 500 and introduced Gaussian observation noise to each species, modeled as $\hat{x}_i(t) = x_i(t) + \varepsilon_i^t$, where $\varepsilon_i^t \sim N(0, e^2)$, with e representing the noise strength. As illustrated in Figs. 4(d), 4(e), and 4(f), InVaXMap maintains a high level of link detection power across a wide range of noise strengths. We further compared the effects of two

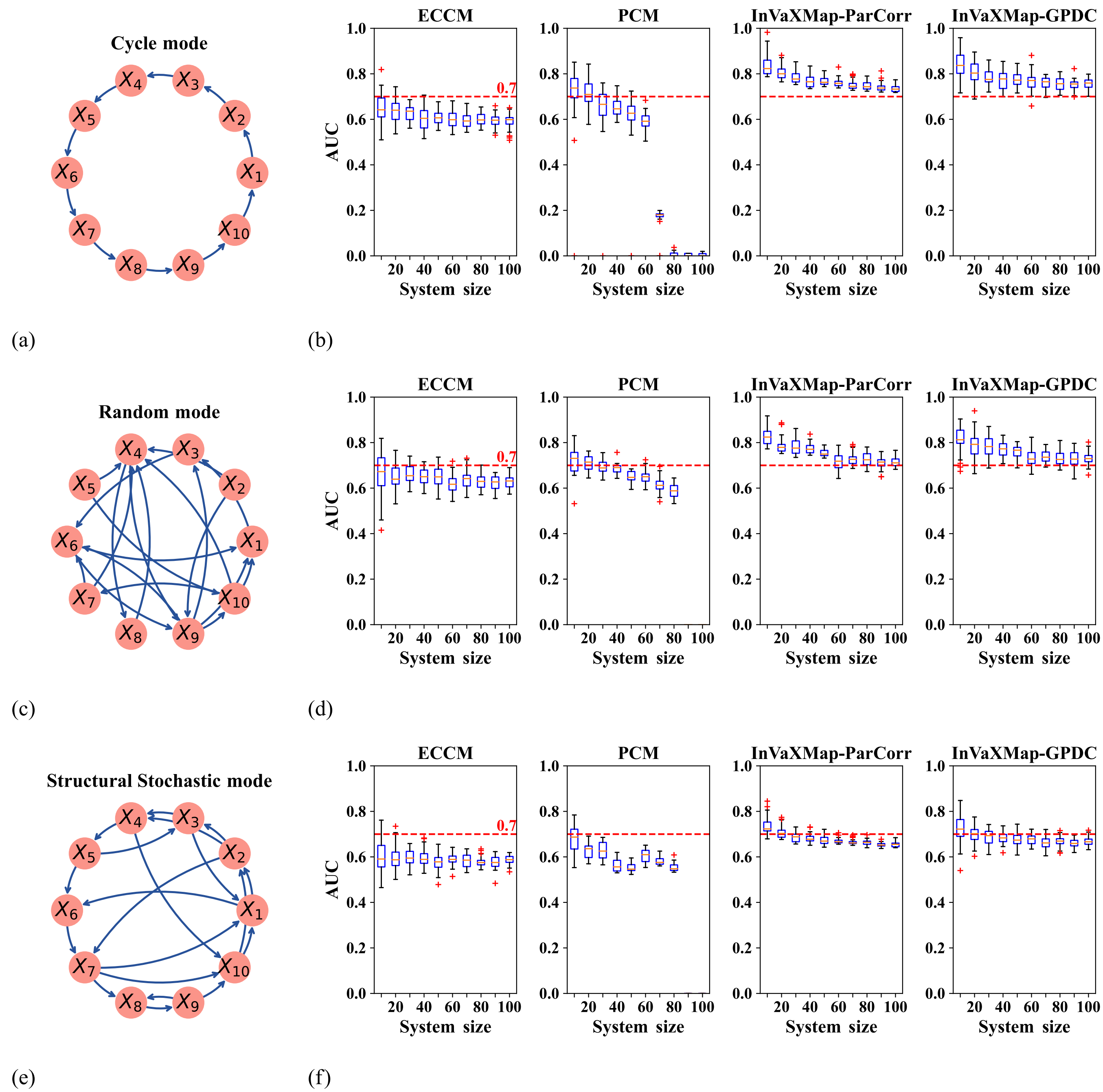

FIG. 3. Physical link inference for benchmark systems using ECCM, PCM, and InVaXMap (with ParCorr and GPDC). (a) Cycle mode. (c) Random mode. (e) Structural stochastic mode. (b),(d),(f) The physical link detection performance of each method across different system sizes. In all scenarios, InVaXMap consistently outperforms ECCM and PCM. The red dashed line at 0.7 serves as a visual reference threshold, though AUC performance varies across systems and configurations.

types of random noise, observation noise and process noise, under different strengths (Appendix E 4). We randomly selected the parameters of benchmark systems within the corresponding intervals for one independent realization, then demonstrated the robustness of InVaXMap across different realizations from the same model through multiple independent realizations for each noise level, as shown in Appendix E 4, Fig. 15. Additionally, the performance of InVaXMap is less affected by the process noise, but more affected by the observation noise, because strong observation noise destroys the structure of the shadow manifold.

We also validated the InVaXMap method using two additional benchmark systems, the multispecies Ricker model and coupled Lorenz systems, in which the interaction

TABLE II. Average AUC of eleven methods on cycle mode.

	System size									
Methods	10	20	30	40	50	60	70	80	90	100
GC	0.478	0.484	0.486	0.486	0.486	0.485	0.485	0.486	0.486	0.485
CGC	0.665	0.498	0.498	0.480	0.488	0.471	0.495	0.485	0.484	0.484
PGC	0.656	0.537	0.521	0.499	0.485	0.485	0.496	0.488	0.491	0.123
TE	0.670	0.650	0.667	0.657	0.662	0.658	0.669	0.667	0.669	0.667
FullCI-ParCorr	0.571	0.492	0.499	0.511	0.522	0.513	0.521	0.516	0.526	0.514
PCMCI-ParCorr	0.622	0.642	0.621	0.604	0.615	0.600	0.609	0.598	0.603	0.595
PCMCI-GPDC	0.603	0.608	0.602	0.593	0.605	0.600	0.597	0.597	0.594	0.592
PCMCI-CMI	0.438	0.451	0.451	0.457	0.451	0.459	0.462	0.465	0.463	0.460
ECCM	0.644	0.638	0.626	0.604	0.607	0.603	0.595	0.602	0.595	0.590
PCM	0.715	0.709	0.648	0.653	0.601	0.593	0.166	-	-	-
InVaXMap	0.836	0.807	0.784	0.768	0.767	0.761	0.751	0.748	0.741	0.735

TABLE III. Average AUC of eleven methods on random mode.

	System size									
Methods	10	20	30	40	50	60	70	80	90	100
GC	0.437	0.482	0.466	0.476	0.479	0.488	0.479	0.490	0.489	0.488
CGC	0.701	0.500	0.511	0.486	0.481	0.493	0.483	0.479	0.489	0.489
PGC	0.688	0.540	0.533	0.490	0.485	0.489	0.486	0.493	0.492	0.119
TE	0.537	0.566	0.557	0.563	0.557	0.557	0.560	0.564	0.554	0.561
FullCI-ParCorr	0.580	0.527	0.516	0.516	0.537	0.529	0.529	0.525	0.530	0.522
PCMCI-ParCorr	0.648	0.622	0.624	0.620	0.617	0.608	0.608	0.604	0.595	0.604
PCMCI-GPDC	0.553	0.601	0.596	0.582	0.578	0.597	0.580	0.587	0.585	0.581
PCMCI-CMI	0.485	0.472	0.474	0.483	0.481	0.477	0.485	0.490	0.477	0.501
ECCM	0.664	0.652	0.663	0.649	0.647	0.619	0.638	0.629	0.623	0.628
PCM	0.717	0.711	0.691	0.686	0.650	0.650	0.614	0.586	-	-
InVaXMap	0.828	0.785	0.783	0.775	0.755	0.717	0.727	0.726	0.711	0.715

TABLE IV. Average AUC of eleven methods on structural stochastic mode.

	System size									
Methods	10	20	30	40	50	60	70	80	90	100
GC	0.459	0.477	0.486	0.480	0.489	0.486	0.483	0.484	0.484	0.483
CGC	0.631	0.496	0.505	0.491	0.492	0.490	0.488	0.489	0.486	0.486
PGC	0.616	0.509	0.518	0.505	0.491	0.486	0.487	0.493	0.488	0.124
TE	0.597	0.612	0.603	0.606	0.611	0.609	0.614	0.621	0.608	0.613
FullCI-ParCorr	0.555	0.510	0.512	0.514	0.517	0.509	0.514	0.511	0.515	0.514
PCMCI-ParCorr	0.594	0.602	0.590	0.588	0.585	0.592	0.586	0.585	0.580	0.581
PCMCI-GPDC	0.550	0.569	0.581	0.578	0.577	0.566	0.574	0.574	0.565	0.570
PCMCI-CMI	0.409	0.435	0.441	0.431	0.446	0.444	0.449	0.445	0.452	0.455
ECCM	0.598	0.595	0.596	0.596	0.578	0.589	0.585	0.578	0.578	0.587
PCM	0.678	0.629	0.629	0.563	0.553	0.606	0.584	0.556	-	-
InVaXMap	0.733	0.706	0.687	0.682	0.672	0.669	0.664	0.663	0.653	0.655

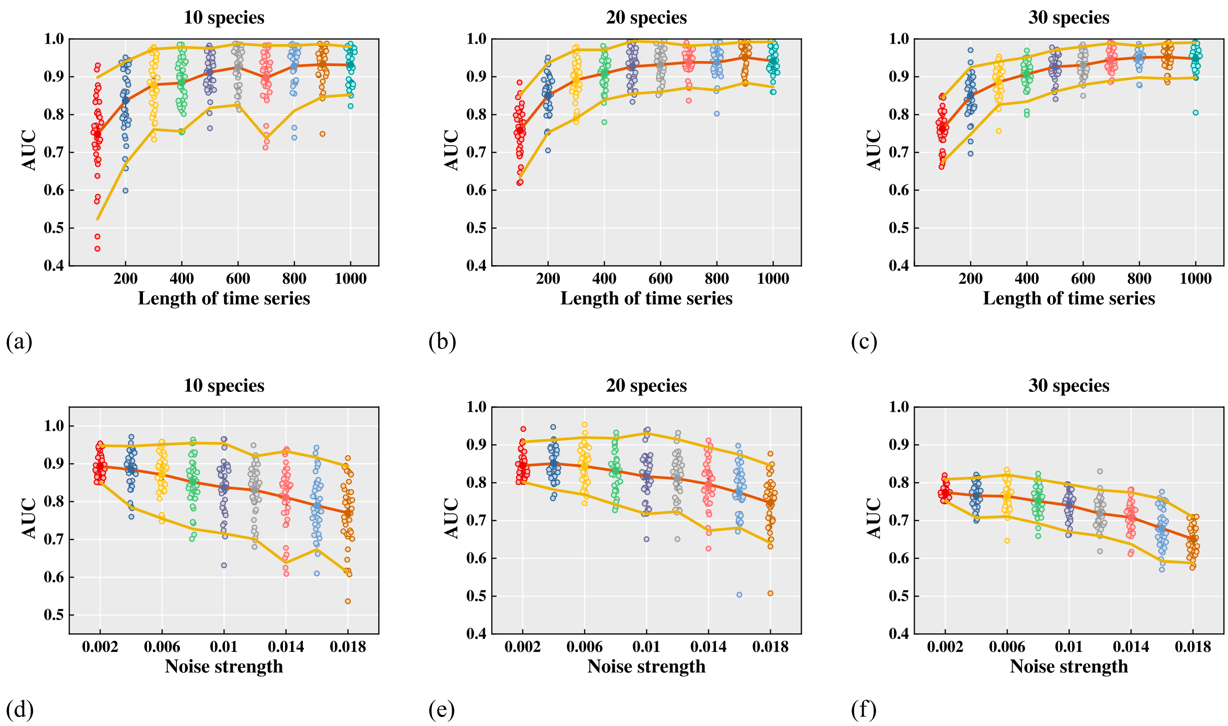

FIG. 4. Robustness of InVaXMap against different length of time series and observation Gaussian noise with different noise strengths. (a)–(c) Link inference performance of InVaXMap versus the length of time series with 10 species (a), 20 species (b), and 30 species (c), respectively. (d)–(f) Link inference performance of InVaXMap versus observation Gaussian noise strength with 10 species (a), 20 species (b), and 30 species (c), respectively. Each data point is averaged over 40 independent realizations. The red lines represent the mean of AUC and the yellow lines represent the 5% and 95% quantiles of AUC.

strengths and network are known *a priori*. The multispecies Ricker model with n species is described as

$$N_i(t+1) = N_i(t)\exp\left[r_{0i}(1 + e_i\mathbf{M}\mathbf{N}(t))\right], \quad i = 1, \ldots, n, \tag{11}$$

where r_{0i} is the intrinsic growth rate of species i, e_i is a vector taking zero values for all except for the ith entity, $\mathbf{M} \in R^{n\times n}$ represents the interaction matrix, and $\mathbf{N}(t) = [N_1(t), \ldots, N_n(t)]^T$. The n coupled Lorenz systems [30] can be described as

$$\begin{aligned}
\dot{x}_k &= \sigma(x_k - y_k) + \varepsilon^t_{k,x}, \\
\dot{y}_k &= \rho x_k - y_k - x_k z_k + \sum_{l=1}^{n} a_{kl}(x_l - y_l) + \varepsilon^t_{k,y}, \\
\dot{z}_k &= \beta z_k + x_k y_k + \varepsilon^t_{k,z},
\end{aligned} \tag{12}$$

where $\sigma = -10$, $\rho = 28$, $\beta = -8/3$, a_{kl} is the connection from lth system to kth system. All parameter settings are detailed in the Appendix. The InVaXMap method outperforms other approaches across different system sizes, as demonstrated in Appendix E 8, Fig. 20.

B. Detecting physical links in real-world systems

We analyzed brain systems using brain activity data obtained from the Neurodynamics Laboratory at the State University of New York Health Center [23]. Ten subjects were selected for the study, and we employed the InVaXMap method to measure the physical link matrix of 35 electrodes positioned in the frontal, parietal, and occipital regions (detailed in Appendix F 1). To examine the influence of alcohol on physical links between electrodes, we conducted controlled experiments, comparing the brain activity of subjects under the influence of alcohol with that of the control group, as illustrated in Figs. 5(a) and 5(b). Our findings indicate a significant reduction in the strength of physical links within and between different brain regions in the alcohol-affected brain. This reduction can be attributed to alcohol impairing the brain's regulatory functions both within and across regions [31,32]. In the control group, we observed numerous bidirectional causal pathways between the frontal and parietal regions, which were largely absent in the alcohol group. The frontal region is primarily responsible for motor functions, while the parietal region is involved in spatial perception and navigation. This aligns with the well-documented motor impairments and sensory deficits associated with alcohol consumption [33–35]. Interestingly, some electrodes, such

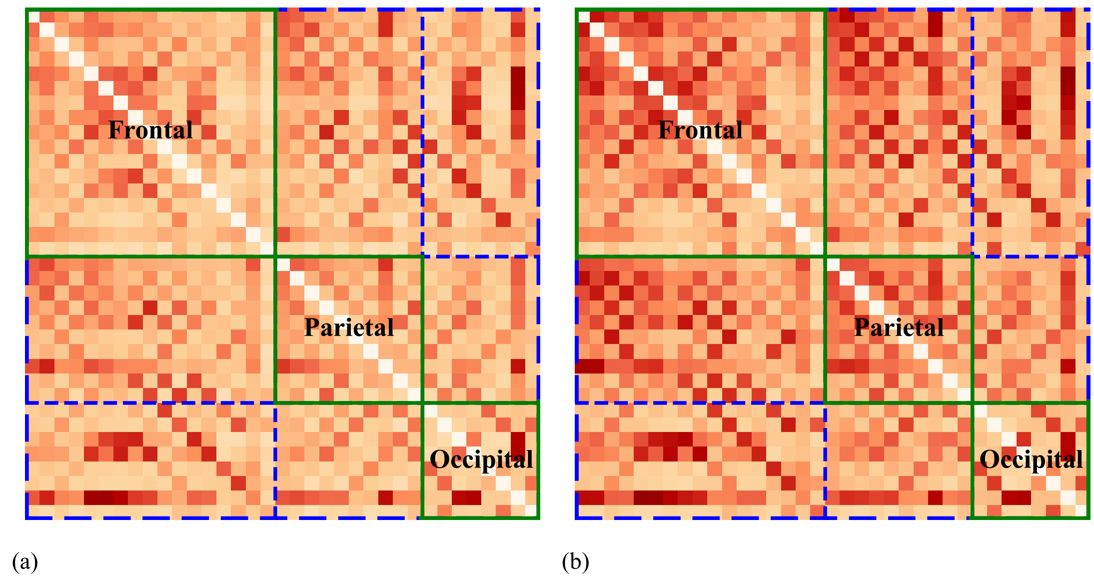

FIG. 5. Link detection in brain systems for the alcoholic brain (a) and the control brain (b). The darker the color, the greater the average causal effect strength. For alcoholic brain, we find substantially reduced causal effect strengths within and between different brain regions. This effect is attributed to the fact that alcohol leads to a decline in control (see Appendix). The green boxes represent the causal effects within each region, while the black dashed lines represent the boundaries between different regions.

as electrode O2 in the occipital region, retained their original physical influence despite the effects of alcohol.

Next, we tested gene regulatory systems using the DREAM4 gene expression dataset from the computational network challenge [24,36,37]. This dataset contains five gene regulatory networks. From each network, we selected 40 interacting genes and combined all instances into a single time series for state-space reconstruction (detailed in Appendix F 2). We then compared the physical relationships identified by InVaXMap with the known physical structures of the five networks and calculated ROC (receiver operating characteristic) curves for each network [Fig. 6(a)]. The average AUC (area under the ROC curve) across the five networks was 0.715, indicating a high degree of accuracy in detecting gene causal interactions. Additionally, competing methods were also applied to test physical relationships in these five selected networks, with InVaXMap achieving the highest AUC in all five networks [Fig. 6(a) and Appendix F 2 Fig. 24]. Figure 6(b) illustrates the relationship between threshold and detection accuracy

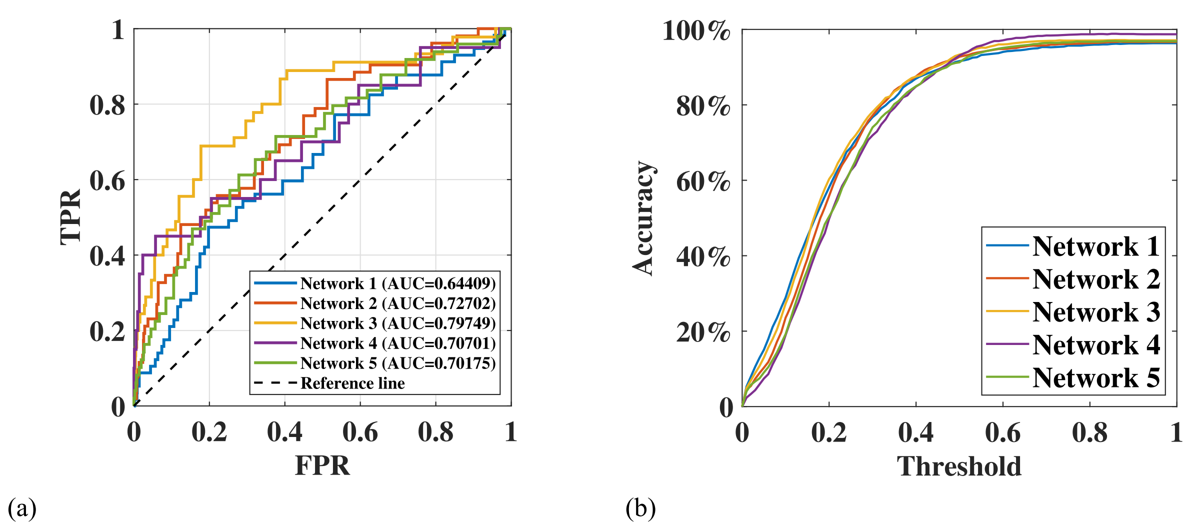

FIG. 6. Link detection in gene regulatory systems. (a) The ROC curves for physical link detection in five representative networks based on InVaXMap. The average area under the ROC curve (AUC) for the five networks was 0.715, which indicates a high level of accuracy in identifying gene physical interactions. (b) The causality detection accuracy versus the threshold. It shows that InVaXMap exhibits a high link detection power within a wide threshold range.

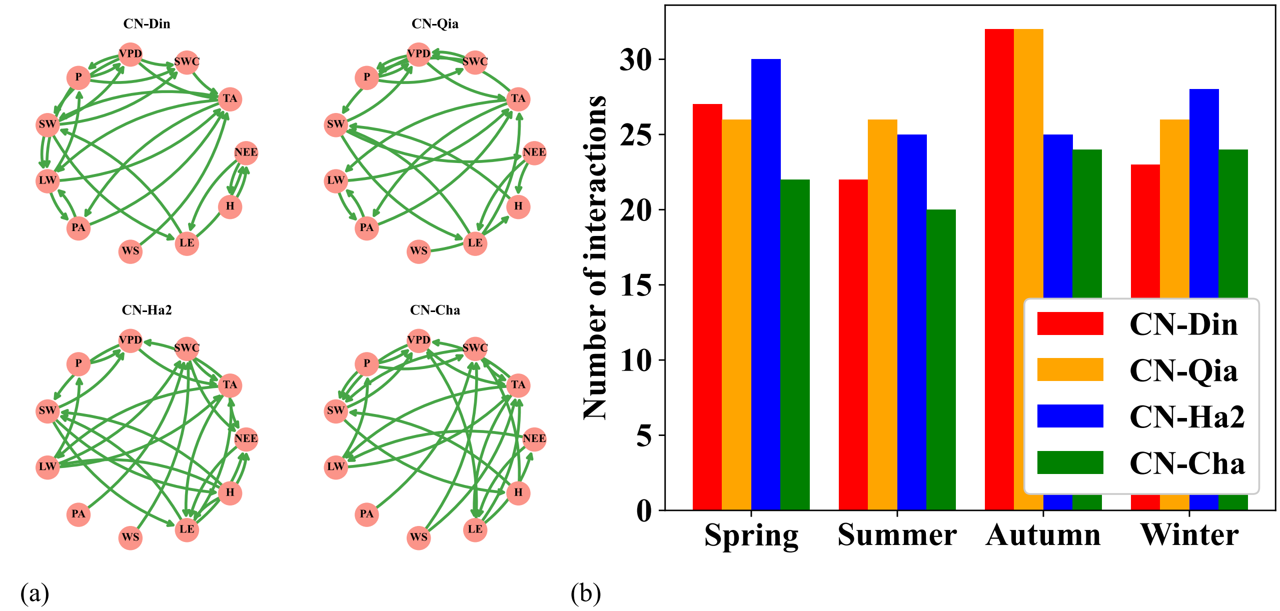

FIG. 7. Link detection in biosphere-atmosphere systems. (a) Physical networks of four selected sites: CN-Din, CN-Qia, CN-Ha2, and CN-Cha, for which we find qualitatively different physical structures. (b) Quantification: Number of physical links in different seasons for each selected site. The number of physical links across the four sites is changing with seasons and reaches the smallest value in summer.

across the five gene regulatory networks, suggesting that the InVaXMap method demonstrates robust physical link detection over a broad range of thresholds.

We also studied biosphere-atmosphere systems using the FLUXNET2015 dataset, which provides ecosystem-scale data from 212 global sites and is widely used in various applications (detailed in Appendix F 3) [25,38,39]. Focusing on four Chinese sites, CN-Din, CN-Qia, CN-Ha2, and CN-Cha, we applied InVaXMap to analyze causal relationships among eleven variables grouped into three categories: net ecosystem exchange, micrometeorology, and energy processing. Our analysis revealed causal relationships among variables influenced by geographical and seasonal factors. As shown in Fig. 7(a), the physical network structure derived from InVaXMap varies with geographic location. The number of physical links detected with coefficients greater than 0.3 across the four sites fluctuates with the seasons, reaching its lowest value during summer [Fig. 7(b)]. For the two specific variables, NEE (net ecosystem exchange) and H (sensible heat flux), we observed that the strength of their physical link changes with both geographical location and season. In lower-latitude sites, CN-Din and CN-Qia, the physical link between NEE and H was stronger in spring and winter compared to summer and autumn (Appendix F 3, Fig. 26 and 27). However, this seasonal variation was not observed at the higher-latitude sites, CN-Ha2 and CN-Cha (Appendix F 3, Fig. 28 and 29). These findings align with previous studies [25,38–43], demonstrating the ability of InVaXMap to detect physical interactions between ecological and atmospheric variables and to capture how these interactions vary across regions and seasons.

Additionally, we carried out experiments on one synthetic gene circuit, where the true wiring is known [26]. The performance of InVaXMap maintained a high level across all tested systems. We also applied the InVaXMap method to analyze the physical relationship between daily air pollutant concentrations and cardiovascular disease hospitalizations (Appendix G, Fig. 31).

These results highlight the broad applicability of InVaXMap to real-world datasets and its potential for reliable modeling and prediction of not only mesoscale networks but also high-dimensional systems.

IV. DISCUSSION

Insight into physical interactions has emerged from two primary strands of modern science: observational discoveries and carefully designed experiments that intervene in the system under well-controlled conditions [44]. However, interventions are often impractical or ethically infeasible to implement [5,45]. Consequently, data-driven approaches based on time series data from simulations or real-world observations have gained significant attention.

We introduced InVaXMap to identify physical cause-effect relations in nonlinear networked systems. A key feature is that it selects relevant variables while accounting for temporal delay effects. This strengthens physical link detection and reduces false positives. Extensive tests on benchmark datasets and real-world systems demonstrate that InVaXMap reliably uncovers physical links. Our findings offer valuable insights into the dynamics of complex systems, including human brain networks, gene regulatory systems, and biosphere-atmosphere interactions, and may prove useful in advancing their understanding and management.

In human brain networks of alcoholic patients, we observed a significant reduction in physical interactions within and between brain regions, consistent with prior research [31,32]. In biosphere-atmosphere systems, we identified a nuanced spatiotemporal effect: the number of physical links across four study sites was at its lowest during the summer. Specifically, the physical link strength between NEE (net ecosystem exchange) and H (sensible heat flux) was higher in the spring and winter compared to summer and autumn at lower-latitude sites, while this seasonal variation was absent at higher-latitude sites. These findings underscore the potential of InVaXMap for reconstructing high-dimensional physical networks (tested up to $n = 100$).

InVaXMap offers two major advantages. First, it effectively handles strongly coupled, nonseparable systems. This remains a challenge for many existing methods, even those that perform well in high-dimensional settings [16,17]. Their limitations become especially apparent when nonseparability arises under weak or moderate coupling strengths.

Second, physical link detection often involves a fundamental trade-off. Including more variables can strengthen the robustness of causal interpretation. At the same time, it can reduce detection power by increasing dimensionality and diluting effect sizes. InVaXMap addresses this problem by selecting only relevant variables during the conditioning stage. It filters out spurious information while retaining direct physical influences. By removing irrelevant variables, InVaXMap reduces the risk of missing significant physical links. This makes it particularly effective in high-dimensional settings.

Our method outperforms established causal graph techniques such as PC and PCMCI. These methods rely on sequential conditional independence testing. In contrast, we use state-space reconstruction. This allows InVaXMap to achieve superior performance, especially in large systems with nonseparable variables.

In many dynamical systems around us, physical effects tend to propagate over time in proportion to their structural path length or physical interaction delay. This principle implies that *direct interactions typically exert their influence on a target variable with shorter time lags* than indirect interactions, which must transmit their effect through intermediate variables. In ecological networks, for instance, a predator's direct impact on its prey manifests quickly, whereas indirect effects through additional trophic levels accumulate over time. Similarly, in neural systems, synaptic transmission between directly connected regions usually introduces shorter delays than polysynaptic or modulatory pathways. Gene regulatory networks also often exhibit hierarchical activation cascades, where primary transcription factors regulate targets more rapidly than downstream effectors influenced via intermediate nodes. These structural and dynamical features justify InVaXMap's assumption that *relevant mediators will typically appear with smaller delays*. This allows reliable and efficient incremental selection of conditioning variables, which suppresses spurious physical links without overconditioning on the full network state.

However, in certain systems such as feedback inhibited gene circuits, hybrid neural hormonal pathways, trophic cascades in ecology, or engineered systems with predictive feedforward control, *indirect effects may manifest with shorter delays than direct physical interactions*. In such cases, the assumption that delay magnitude aligns with physical immediacy breaks down. For example, a direct transcriptional effect from gene A to gene C might require a long processing time, while an indirect enzymatic path via gene B modulates C far more rapidly. In ecological systems, intermediate predators may respond swiftly to top down pressure, influencing lower trophic levels before slower direct interactions take effect. Similar patterns occur in robotic systems where anticipatory modules (e.g., $A \rightarrow B \rightarrow C$) stabilize outputs faster than delayed feedback ($A \rightarrow C$). In such scenarios, InVaXMap's delay-ordered conditioning may mistakenly prioritize indirect mediators and filter out true direct links. This can occur particularly when partial correlation removes residual influence following early conditioning (Fig. 8). While these cases may not dominate in typical natural systems, they represent meaningful exceptions and motivate further refinement of the delay-based selection heuristic. However, InVaXMap does not universally fail in scenarios where indirect effects manifest with shorter delays than direct physical interactions. The method remains valid when the direct interaction does not share identical information with the indirect interaction about the target variable, even in cases where the indirect interaction occurs on shorter timescales than the direct one (Fig. 9). Nonetheless, we propose two diagnostic signals to flag potential failure cases: (i) pronounced symmetry in the reconstructed manifold, and (ii) a link-prediction coefficient that does not remain consistent as the time-series length increases. In addition, incorporating a small number of prior-known edges can substantially improve InVaXMap's link predictions (Fig. 10). Further details are provided in Appendix C.

InVaXMap introduces a novel approach to direct link inference in nonlinear dynamical systems by combining *extended convergent cross mapping* with a *delay ordered, incremental construction of the conditioning set*. Traditional cross-mapping methods suffer from interaction transitivity and overcondition on irrelevant variables. Conditional independence approaches, in turn, often do

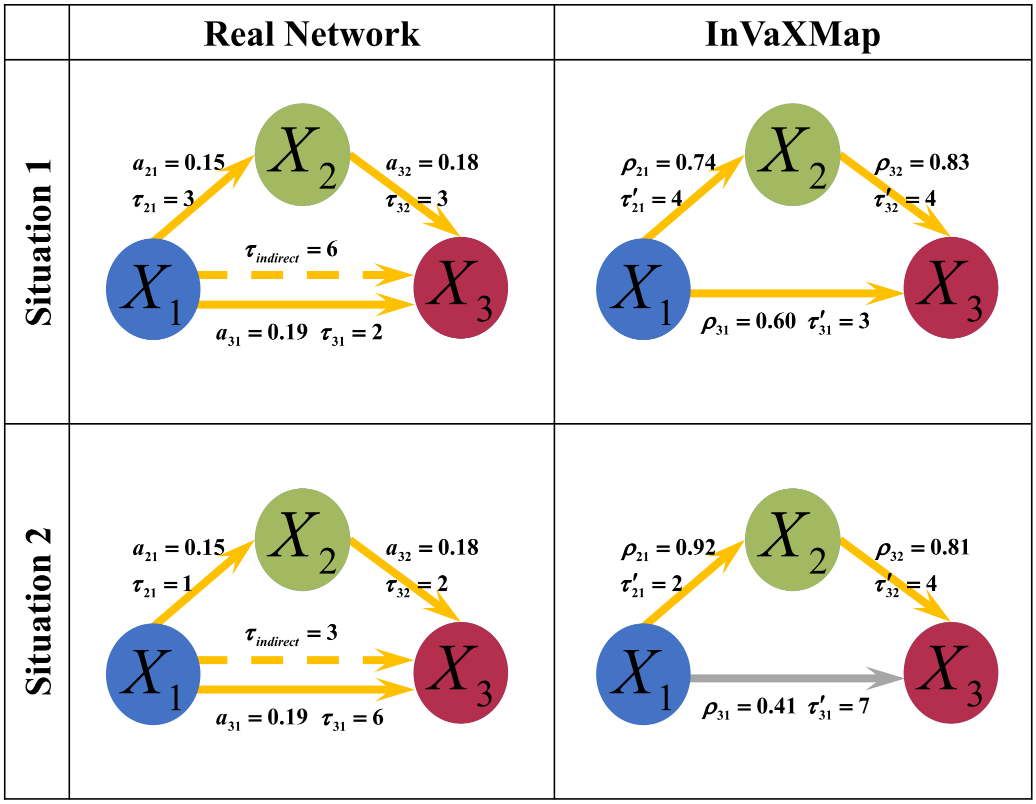

FIG. 8. Supporting experiment on incremental variable construction. We study a three-species discrete Lotka-Volterra competition model with three direct physical links: $X_1 \to X_2$, $X_2 \to X_3$, and $X_1 \to X_3$. There are both direct and indirect physical links from X_1 to X_3. When the time delay of indirect physical link is longer than that of direct physical link (situation 1), InVaXMap correctly detects the real network. When the time delay of indirect physical link is shorter than that of direct physical link (situation 2), InVaXMap's delay ordered conditioning can mistakenly prioritize indirect mediators and filter out true direct causes. The left column presents parameters as defined in the model and the right column presents those inferred by InVaXMap. a_{ij} is the interaction strength from X_j to X_i and τ_{ij} is the corresponding time delay. ρ_{ij} is the physical relationship detected by InVaXMap using the standard partial correlation coefficient and τ'_{ij} is the corresponding time delay. The gray line represents the physical link that InVaXMap fails to detect.

not scale well to high-dimensional systems. In contrast, InVaXMap strategically constructs a compact, physically informative variable subset by exploiting temporal delay structure. This ordering is then used to *filter indirect effects via partial correlation*, enabling both precision and scalability. The novelty of InVaXMap lies in this *integration of time delay information as a guiding principle for conditioning*, a strategy that effectively balances the trade off between physical completeness and statistical efficiency. By doing so, it achieves superior robustness in systems with *nonseparable, weakly coupled variables*, a regime where other methods often fail. To our knowledge, no existing method combines these elements in the same structured and efficient manner.

Prior work [46] proposes a causal discovery method based on structural causal models, which infer causal relationships by finding a topological ordering of variables and then refining the causal graph using conditional independence tests. Their approach assumes that causal influences are instantaneous or separable and does not account for temporal dependencies. InVaXMap, by contrast, is specifically designed for time-series dynamical systems, which reconstructs physical links from observed trajectories using state-space reconstruction and ECCM to capture time-lagged effects. A key computational advantage of InVaXMap is its incremental variable construction, which selectively builds the conditioning set instead of integrating all variables at once. This not only reduces time complexity in computing partial correlations but also ensures numerical stability in cases where the number of conditioning variables exceeds the time series length, avoiding issues where standard partial correlation coefficients become unstable.

There remain limitations and opportunities for future research. First, InVaXMap is most effective when applied to datasets generated by deterministic nonlinear systems governed by underlying state equations. While we have demonstrated its capability in handling both weakly and strongly coupled systems, further improvements could enhance its performance in dealing with generalized synchronization in such environments. Second, InVaXMap requires sufficiently long time series data and appropriate observation noise levels for accurate physical link discovery. Both factors influence the accuracy of InVaXMap in inferring physical relationships (Fig. 4 and Appendix E 4, Fig. 15) since an unreliable reconstruction of the shadow manifold will degrade performance. Extending its applicability to high-dimensional datasets with shorter time series, possibly through the integration of machine learning techniques, represents a promising avenue for future research [47]. Third, the effectiveness of InVaXMap depends on an appropriate time scale, as the time step of the time series should be comparable to the system's dynamic operation. Insufficient sampling rates can lead to inaccurate physical link inference. Finally, like many other methods, InVaXMap assumes that all variables in the system are observed and that interactions, along with their time delays, remain constant over time. The presence of unobserved variables or dynamically evolving interactions may impair its performance. While related methods have been proposed to address this issue [11,48], further exploration is needed.

In summary, InVaXMap fundamentally enhances the physical link inference accuracy compared to existing approaches, providing a robust framework for identifying fundamental physical cause-effect relations in complex networked systems. While our method builds on well-established techniques such as Takens' embedding, extended convergent cross mapping, and partial correlation, its key strength lies in the incremental selection mechanism that constructs a well-suited conditioning set. This simple yet unexpectedly effective, physics-informed strategy distinguishes our approach from existing methods.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The data that support the findings of this article are openly available [23, 24, 25, 27].

APPENDIX A: CONCEPT OF NONSEPARABILITY

Under mild conditions, mainly, that the system's evolution is a smooth map on a compact state space, Takens' delay embedding theorem guarantees that we can recover the entire state trajectory from just one observable. In practice, this means that by stacking time delayed samples of a single variable, for example, $x(t)$, $x(t-\tau)$, $x(t-2\tau)$, …, we obtain a one to one reconstruction of the original dynamics. A single suitably chosen measurement carries all of the information needed to rebuild the full system. This property—nonseparability—means that one cannot truly isolate one variable's information from the rest when you want to predict a nonlinear system's future behavior.

Precisely, start with a continuous-time dynamical system of n variables of the following general form

$$\dot{\mathrm{x}} = F(\mathrm{x}), \tag{A1}$$

the state variable $x(t) = [x_1(t), x_2(t), \ldots, x_n(t)]^T$ evolves inside a compact manifold M_x. Based on Takens' embedding theory [7,8], $\Gamma_{h,\varphi,\tau}(\mathrm{x}) = [h(\mathrm{x}), h(\varphi_{-\tau}(\mathrm{x})), \ldots, h(\varphi_{-(L-1)\tau}(\mathrm{x}))]^T$ is generically an embedding map, where φ_t is a flow along the manifold M_x and $h\colon M_\mathrm{x} \to R$ is a smooth observation function. Here we take the observation function $h(\mathrm{x}) = x_i$, where x_i is the ith component of x. Then $\mathrm{y}(t) = [x_i(t), x_i(t-\tau), \ldots, x_i(t-(L-1)\tau)]^T$ forms a shadow manifold M_y. The dynamics ψ_t on M_y are topologically conjugated with the dynamics φ_t on M_x,

$$y(t+\tau) = \psi_\tau(y(t)) = \Gamma \circ \varphi_\tau \circ \Gamma^{-1}(y(t)). \tag{A2}$$

From Eq. (A1), the mapping $[\varphi_\tau]_j\colon x(t) = [x_1(t), x_2(t), \ldots, x_n(t)]^T \to x_j(t+\tau)$ indicates the dynamics of the component x_j is governed by $x_1(t), x_2(t), \ldots, x_n(t)$. On the other hand, the dynamics of the component x_j is governed by $[\Gamma^{-1}\psi_\tau]_j\colon y(t) = [x_i(t), x_i(t-\tau), \ldots, x_n(t-(L-1)\tau)]^T \to x_j(t+\tau)$.

APPENDIX B: SCIENTIFIC GAP BETWEEN ECCM AND InVaXMap

Link inference in complex systems becomes particularly challenging when the variables of the underlying dynamics are *nonseparable*, which results in a state-space reconstructions overlap, preventing a clear distinction between individual influences. To address such challenges, *state-space reconstruction methods* have been developed to infer causality in systems where traditional separability-based approaches fail.

One such approach, *convergent cross mapping*, has been widely used for causal inference in dynamical systems. While originally developed to address challenges in nonseparable systems, its effectiveness may still be constrained in scenarios with strong variable interdependencies or when the assumptions underlying state space reconstruction are violated [15].

To overcome some of these limitations, multiview distance-regularized S map was introduced to reconstruct high-dimensional interaction Jacobian networks without relying on explicit model assumptions [17]. However, identifying direct physical interactions in high-dimensional nonlinear systems, where variables are inherently dependent and thus nonseparable, remains an open challenge.

ECCM has been shown to be effective in detecting causality even when system variables are nonseparable, particularly in low-dimensional nonlinear systems.

However, it struggles to differentiate between true causal relationships and spurious links introduced by causation transitivity or common drivers. For example, while ECCM successfully identified the optimal cross-map lags and cross-map skill to differentiate direct from indirect effects in a system structured as $X_1 \to X_2 \to X_3 \to X_4$, the high variance in cross-map skill measures indicated reduced reliability in distinguishing direct causation [29]. These limitations become even more pronounced in high-dimensional nonlinear systems.

To overcome these challenges, we developed InVaXMap, a novel method that leverages generalized Takens' theorems to infer direct physical links by selectively retaining only the most relevant variables. By doing so, InVaXMap effectively filters out spurious physical links caused by transitivity or common drivers, issues that ECCM typically fails to address. As a result, InVaXMap effectively eliminates spurious physical links caused by transitivity or common drivers, which ECCM often fails to filter out. This results in more accurate link detection in high-dimensional nonlinear systems while also mitigating the curse of dimensionality, as demonstrated by simulation experiments.

APPENDIX C: ADDITIONAL INFORMATION ABOUT InVaXMap

1. Justification of the central physical intuition

The central physical intuition behind InVaXMap is that direct effects typically manifest at shorter delays than indirect (cascade) effects. When this ordering is violated—for example, in certain feedforward motifs where an indirect pathway acts faster than the direct one—the intuition may break down. However, InVaXMap does not necessarily fail in all such cases. The method can still succeed when the direct interaction contributes information about the target that is not redundant with the indirect pathway, even if the indirect effect occurs at a shorter delay.

To quantify this behavior, we performed controlled tests in two settings where indirect interactions act on shorter

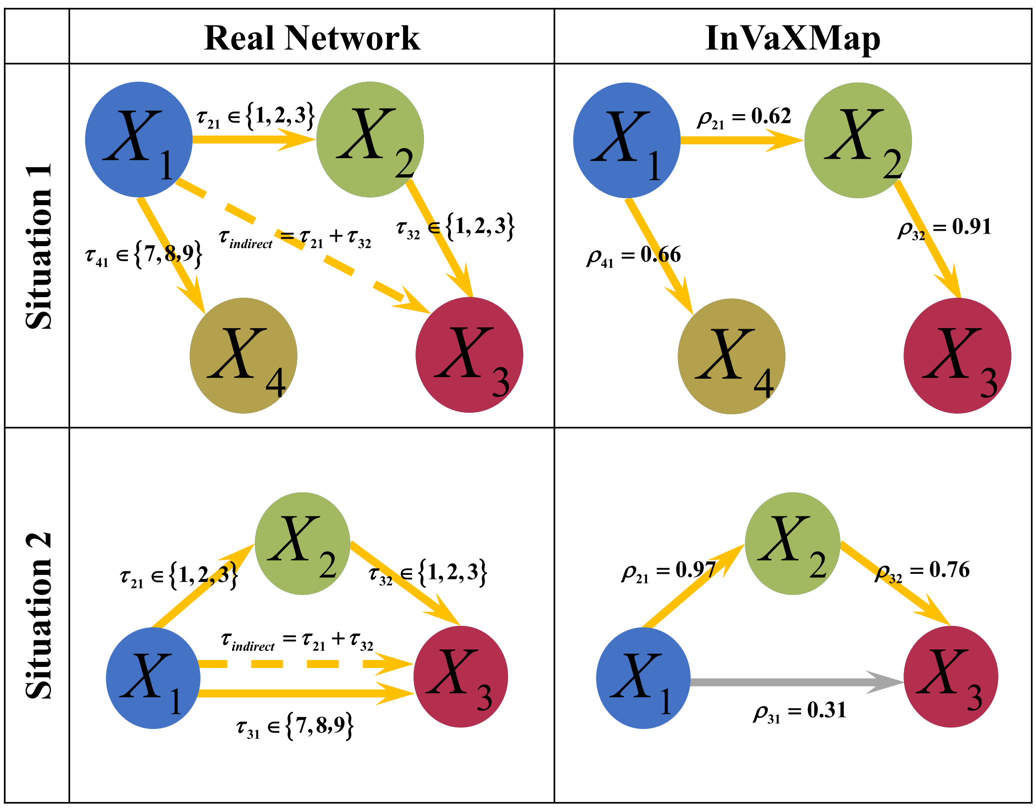


FIG. 9. Controlled validation of the delay-ordered incremental variable construction (IVC) used in InVaXMap on a multispecies discrete Lotka-Volterra competition model. *Left:* ground-truth motifs with imposed interaction delays τ_{ji} for $X_i \to X_j$; dashed arrows denote an indirect influence with total delay $\tau_{\text{indirect}} = \tau_{21} + \tau_{32}$. *Right:* corresponding networks inferred by InVaXMap, annotated by the link scores ρ_{ji}. We consider two cases in which an indirect pathway acts on a shorter time scale than a direct interaction. Situation 1 (distinct pathways): the indirect cascade $X_1 \to X_2 \to X_3$ (with $\tau_{21}, \tau_{32} \in \{1,2,3\}$) is faster than the unrelated direct link $X_1 \to X_4$ (with $\tau_{41} \in \{7,8,9\}$); because the indirect and direct effects pertain to different targets, InVaXMap correctly recovers the true direct edges. Situation 2 (same target/feed-forward loop): the indirect pathway $X_1 \to X_2 \to X_3$ is faster than the direct link $X_1 \to X_3$ (with $\tau_{31} \in \{7,8,9\}$); delay-ordered conditioning may then prioritize the indirect mediator and suppress the true direct cause (gray arrow, low ρ_{31}).

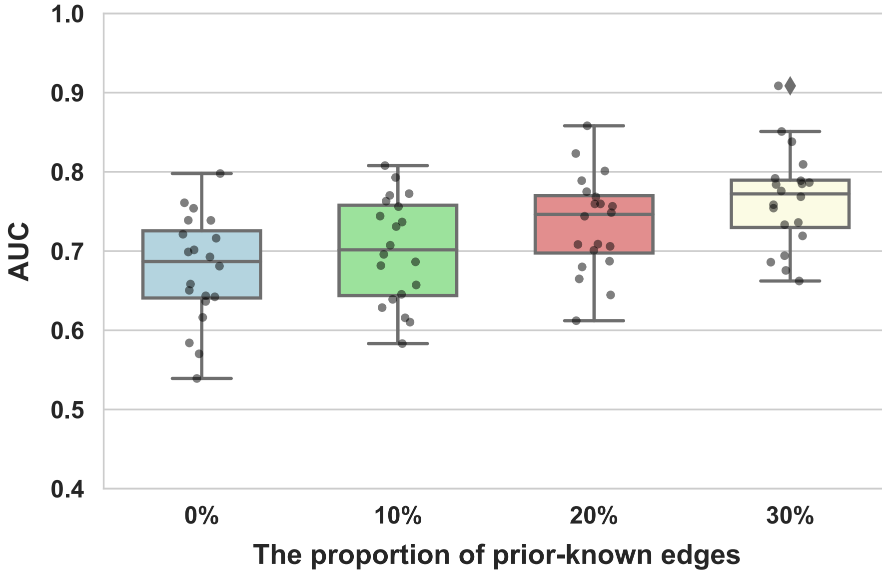


FIG. 10. Performance of InVaXMap with different proportions of prior-known edges. The performance of InVaXMap consistently improves as the fraction of prior-known edges increases, demonstrating that incorporating prior knowledge enhances the accuracy and robustness of the method.

time scales than direct ones. For each link, we randomly assigned a delay and averaged results over 20 independent runs (Fig. 9). In *situation 1*, the directly and indirectly affected variables belong to distinct physical pathways, so the direct interaction does not convey the same information about the target as the indirect pathway. In this case, InVaXMap correctly recovers the true physical relations, even when a mediator has a longer delay than an irrelevant variable. In *situation 2*, the direct and indirect effects correspond to the same variable, so the direct interaction is informationally redundant with the indirect pathway. Here, the delay-ordered conditioning in InVaXMap may prioritize indirect mediators and mistakenly discard true direct links.

To help detect such failure cases, we propose two diagnostic signals. First, pronounced symmetry in the manifold reconstructed from the target (relative to that from the response) may indicate a breakdown of the underlying assumption. Second, the link-prediction coefficient should remain stable as the time-series length increases; strong sensitivity to length suggests unreliable inference. A systematic study of the validity and generality of these indicators is beyond the scope of this work given the resubmission timeline. The development of robust diagnostics therefore remains an important direction for future research.

Finally, prior knowledge about network edges—especially those acting on long timescales—can improve InVaXMap's accuracy. If an edge $X \to Y$ is known, then once Y is selected by the delay-ordering step, it is added to I_a without requiring a partial-correlation test. We validated this strategy on a 20-species discrete Lotka-Volterra competition model. In each of 20 independent experiments, we randomly defined 20 interactions and evaluated different proportions of prior-known edges. The results (Fig. 10) show a consistent performance gain as the proportion of prior-known edges increases, indicating that incorporating prior knowledge enhances both accuracy and robustness.

2. Partial correlation indices used in InVaXMap

For robustness, we consider two versions: (i) InVaXMap with a linear partial correlation index, based on the standard partial correlation coefficient (ParCorr), and (ii) InVaXMap with a nonlinear partial correlation index, based on Gaussian process regression and a distance correlation test (GPDC), which captures more complex dependencies.

Linear partial correlation index—In general, partial correlation measures how much of the relationship between X_1 and X_j remains after controlling for other variables X_k. The linear partial correlation index is defined as the absolute value of the partial correlation coefficient, ensuring that both positive and negative dependencies are captured equally:

$$\rho^{I_a}_{X_1 \to X_j} = \left| \mathrm{pcor}\left(X_1, X_1^{X_j} \middle| \left\{ X_1^{X_k} | X_k \in I_a \right\} \right) \right|. \tag{C1}$$

This index is derived from the standard partial correlation formula:

$$\mathrm{pcor}(X, Y|Z) = \frac{r(X, Y) - r(X, Z) r(Y, Z)}{\sqrt{[1 - r^2(X, Z)][1 - r^2(Y, Z)]}}. \tag{C2}$$

where $r(X, Y)$ is the Pearson correlation coefficient:

$$r(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}. \tag{C3}$$

Nonlinear partial correlation index—To capture nonlinear dependencies, we define a nonlinear partial correlation index based on Gaussian process regression and a distance correlation test (GPDC):

$$\rho^{I_a}_{X_1 \to X_j} = \left| \mathrm{GPDC}\left(X_1, X_1^{X_j} \middle| \left\{ X_1^{X_k} | X_k \in I_a \right\} \right) \right|. \tag{C4}$$

This method consists of

(1) Gaussian process (GP) regression [49]:

$$\hat{X}_1 = \mathrm{GP}\left(X_1^{X_k}\right), \ \hat{X}_1^{X_j} = \mathrm{GP}\left(X_1^{X_k}\right), \ X_k \in I_a. \tag{C5}$$

(2) 2. Distance correlation (DC) test for unconditional independence [50]:

$$\rho^{I_a}_{X_1 \to X_j} = \mathrm{dcor}(e_1, e_2), \qquad e_1 = |X_1 - \hat{X}_1|,$$
$$e_2 = \left| X_1^{X_j} - \hat{X}_1^{X_j} \right|, \tag{C6}$$

where distance correlation is defined as

$$\mathrm{dcor}(X, Y) = \frac{\mathrm{dCov}(X, Y)}{\sqrt{\mathrm{dVar}(X)\mathrm{dVar}(Y)}}. \tag{C7}$$

The distance covariance and variance are computed as

$$\mathrm{dCov}^2(X, Y) = \frac{1}{n^2} \sum_{j=1}^{n} \sum_{k=1}^{n} A_{j,k} B_{j,k}, \tag{C8}$$

where $A_{j,k}$ and $B_{j,k}$ are the double-centered distance matrices of X and Y, respectively, and n is the sample size.

We implemented GPDC using Python. First, GP was conducted with the “sklearn” package, utilizing a combination of kernel “RBF” and “WhiteKernel.” Then, DC was computed via the “dcor” package, leveraging its AVL tree implementation for efficiency.

3. Parameter choices

In this section, we detail the parameter choices for all experiments. The performance of InVaXMap depends critically on several parameters: the max delay $l_{\max}$, embedding dimension E and time lag τ. The embedding dimension E was selected according to previous research [51]. We set time lag $\tau = 1$ for all experiments, which is consistent with other embedding-based methods [15,29]. For Lotka-Volterra model, max delay $l_{\max} = 10$ when system size is lower than 50, and $l_{\max} = 20$ for larger systems. $l_{\max}$ was held fixed in other systems respectively. Above parameters are listed in Table V.

TABLE V. Embedding parameters for main experiments.

	Embedding dimension E	Time lag τ	Max delay $l_{\max}$
Lotka-Volterra model	4	1	10,20
Richer model	4	1	15
Coupled Lorenz	4	1	10
Brain systems	4	1	10
Gene regulatory systems	3	1	5
Biosphere-atmosphere systems	4	1	10
Synthetic gene circuit	3	1	10
Air pollution dataset	6	1	20

4. Threshold selection

The selection of the threshold is empirical, but the robustness tests on the detection accuracy versus different thresholds show that effective threshold value exists in a wide range [Figs. 6(b), 11, and 13(b)]. For practical application, unsupervised classifiers could be applied to determine the threshold value by classifying the detection results into two groups (i.e., with or without causal links). For instance, the threshold automatically induced from the k-means classifier successfully eliminates most indirect links [Fig. 13(b)]. These results provide a robust, data-driven strategy for threshold selection.

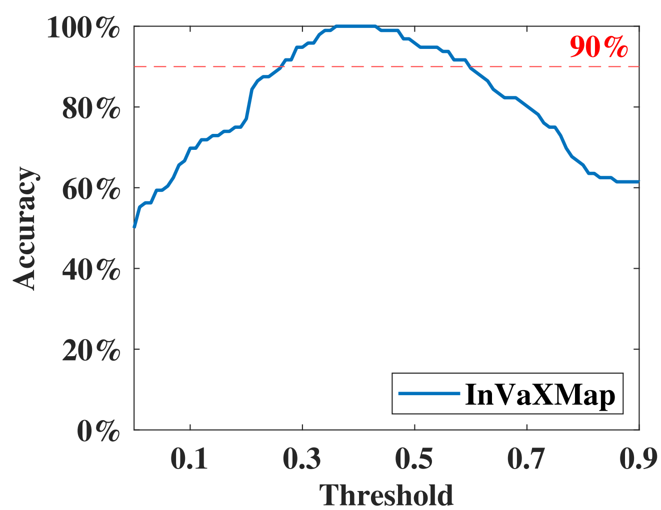


FIG. 11. Link detection accuracy of InVaXMap for all possible interacting modes of three-species systems versus threshold used in InVaXMap. The threshold value at which the accuracy of InVaXMap exceeds 90% ranges from 0.27 to 0.59, covering 32% of the entire threshold range.

APPENDIX D: CONVENTIONAL METHODS USED FOR COMPARISON

1. Partial cross mapping

We begin by introducing the definition of the partial correlation coefficient [52]. Given time series X, Y, and Z, the partial correlation coefficient between X and Y, conditioned on Z, is defined as

$$\mathrm{pcor}(X,Y|Z) = \frac{r(X,Y) - r(X,Z)r(Y,Z)}{\sqrt{(1-r^2(X,Z))(1-r^2(Y,Z))}}, \tag{D1}$$

where $r(X,Y) = \{[\mathrm{Cov}(X,Y)]/\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}\}$ is the correlation coefficient between X and Y.

For time series X, Y, Z_1, and Z_2, the partial correlation coefficient between X and Y, conditioned on both Z_1 and Z_2, is given by

$$\mathrm{pcor}(X,Y|Z_1,Z_2) = \frac{\mathrm{pcor}(X,Y|Z_1) - \mathrm{pcor}(X,Z_2|Z_1)\mathrm{pcor}(Y,Z_2|Z_1)}{\sqrt{(1-\mathrm{pcor}^2(X,Z_2|Z_1))(1-\mathrm{pcor}^2(Y,Z_2|Z_1))}}. \tag{D2}$$

Partial cross mapping (PCM) [9] was developed to eliminate indirect causal influences in systems where non-separability exists among variables. For a complex system with interacting variables X, Y, and Z_i (where $i = 1, 2, \ldots, N-2$), the first-order PCM is defined as

$$\rho^1_{\mathrm{PCM}} = \left|\mathrm{pcor}\left(X, X^Y \middle| \left\{X^{Z_i^Y} \middle| i = 1, 2, \ldots, N-2\right\}\right)\right|.$$

This represents the direct causation from X to Y, conditioned on Z_i (for $i = 1, 2, \ldots, N-2$), where X^Y is the mapping from Y, and $X^{Z_i^Y}$ is the mapping from Z_i^Y, which itself is the mapping from Y. Here, $|\mathrm{pcor}(\bullet)|$ denotes the partial correlation coefficient.

Indirect causation can propagate through multiple variables. The k-order PCM can be defined by considering any combination of k (where $k \le N-2$) mediating variables from Z_i, $i = 1, 2, \ldots, N-2$, as

$$\rho^k_{\mathrm{PCM}} = \left|\mathrm{pcor}\left(X, X^Y \middle| \left\{X^{Z_{i_1}{}^{\cdots Z_{i_k}{}^{Y}}} \middle| (i_1, \ldots, i_k) \text{ is a } k \text{ combination from } \{1, 2 \cdots, N-2\}\right\}\right)\right|. \tag{D3}$$

2. Granger causality

Granger causality (GC) [53] measures the causality between variables by testing whether the prediction of X improves after adding Y, which is based on vector autoregression model. If X_t and Y_t are time series, then

$$X_t = \sum_{i=1}^{L} a_{1i} X_{t-i} + \varepsilon_{1t},$$
$$X_t = \sum_{i=1}^{L} a_{2i} X_{t-i} + \sum_{i=1}^{L} b_{2i} Y_{t-i} + \varepsilon_{2t}. \tag{D4}$$

Granger causality from Y to X can be measured by

$$F_{Y\to X} = \ln\left(\frac{|\mathrm{var}(\varepsilon_{1t})|}{|\mathrm{var}(\varepsilon_{2t})|}\right). \tag{D5}$$

Conditional Granger causality (CGC) [54,55] deals with indirect causality by explicitly removing the influence of a third variable and thus avoids producing misleading causality. For three time series X_t, Y_t, Z_t,

$$X_t = \sum_{i=1}^{L} a_{3i} X_{t-i} + \sum_{i=1}^{L} c_{3i} Z_{t-i} + \varepsilon_{3t},$$
$$Z_t = \sum_{i=1}^{L} a_{4i} X_{t-i} + \sum_{i=1}^{L} c_{4i} Z_{t-i} + \varepsilon_{4t},$$
$$X_t = \sum_{i=1}^{L} a_{5i} X_{t-i} + \sum_{i=1}^{L} b_{5i} Y_{t-i} + \sum_{i=1}^{L} c_{5i} Z_{t-i} + \varepsilon_{5t},$$
$$Z_t = \sum_{i=1}^{L} a_{6i} X_{t-i} + \sum_{i=1}^{L} b_{6i} Y_{t-i} + \sum_{i=1}^{L} c_{6i} Z_{t-i} + \varepsilon_{6t}. \tag{D6}$$

Conditional Granger causality from Y to X is defined as

$$F_{Y\to X} = \ln\left(\frac{|\mathrm{var}(\varepsilon_{3t})|}{|\mathrm{var}(\varepsilon_{4t})|}\right). \tag{D7}$$

Partial Granger causality (PGC) [56,57] is an extension of conditional Granger causality, which further considers the influence of environmental inputs and unmeasured latent variables. Let

$$S = \begin{bmatrix} \mathrm{var}(\varepsilon_{3t}) & \mathrm{cov}(\varepsilon_{3t}, \varepsilon_{4t}) \\ \mathrm{cov}(\varepsilon_{3t}, \varepsilon_{4t}) & \mathrm{var}(\varepsilon_{4t}) \end{bmatrix} = \begin{bmatrix} S_{xx} & S_{xz} \\ S_{zx} & S_{zz} \end{bmatrix},$$
$$\sum = \begin{bmatrix} \mathrm{var}(\varepsilon_{5t}) & \mathrm{cov}(\varepsilon_{5t}, \varepsilon_{6t}) \\ \mathrm{cov}(\varepsilon_{5t}, \varepsilon_{6t}) & \mathrm{var}(\varepsilon_{6t}) \end{bmatrix} = \begin{bmatrix} \sum_{xx} & \sum_{xz} \\ \sum_{zx} & \sum_{zz} \end{bmatrix}. \tag{D8}$$

Partial Granger causality from Y to X is defined as

$$F_{Y\to X} = \ln\left(\frac{|S_{xx} - S_{xz} S_{zz}^{-1} S_{zx}|}{|\Sigma_{xx} - \Sigma_{xz} \Sigma_{zz}^{-1} \Sigma_{zx}|}\right). \tag{D9}$$

3. Transfer entropy

Transfer entropy (TE) [58] is a nonparametric statistic that measures the directed transmission of information between two random processes. The transfer entropy from X to Y refers to reduction in uncertainty of Y given the past of X. More specifically, if X_t and Y_t represent two random processes and information is measured by Shannon entropy, then transfer entropy can be written as

$$T_{X\to Y} = H(Y_t|Y_{t-1:t-L}) - H(Y_t|Y_{t-1:t-L}, X_{t-1:t-L}), \quad \text{(D10)}$$

where $H(X)$ is Shannon entropy of X.

4. Conditional independence

Conditional independence is a measurement for causality, which can be measured by different criteria, like partial correlation. Conditional independence of X and Y given Z is denoted as $X \perp\!\!\!\perp Y|Z$. Consider a system with $\{X_1^t, \ldots, X_N^t\}$, full conditional independence (FullCI) [5,16] measures causality by directly testing

$$X_{t-\tau}^i \not\perp\!\!\!\perp X_t^j | X_t^- \setminus \{X_{t-\tau}^i\}; \quad \text{(D11)}$$

the past X_t^- is truncated at a maximum time lag $\tau_{\max}$.

PCMCI [16] is a state-of-art method to estimate causal networks from large-scale stochastic systems, which consists of two stages: PC [59] condition selection to identify relevant conditions $\hat{\mathcal{P}}(X_t^j)$ for all time series variables $X_t^j \in \{X_1^t, \ldots, X_N^t\}$ and then momentary conditional independence (MCI) to test whether $X_{t-\tau}^i \to X_t^j$ with

$$X_{t-\tau}^i \not\perp\!\!\!\perp X_t^j | \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\}, \hat{\mathcal{P}}(X_{t-\tau}^i) \quad \text{(D12)}$$

APPENDIX E: DETAIL AND ADDITIONAL EXPERIMENTS OF LINK DETECTION IN BENCHMARK SYSTEMS

1. Multispecies discrete Lotka-Volterra competition model with random noises

We validated the InVaXMap method using time series generated from the multi-species discrete Lotka-Volterra competition model [17], described as

$$x_i(t+1) = x_i(t)\left(r_i - \sum a_{ij} x_j(t - \tau_{ij})\right) + \eta_i(t), \quad \text{(E1)}$$

where $x_i(t)$ represents the abundance of species i at time t, and r_i is the growth rate of species i, which is uniformly and randomly selected between 3.4 and 3.8. The interaction strength, a_{ij}, when nonzero, is uniformly chosen between 0.2 and 0.4. The time delay, τ_{ij}, is uniformly selected between 1 and 6. The process noise, $\eta_i(t)$, is fixed at 0.01 for all simulations.

We analyzed three distinct modes of physically interacting systems: cycle mode, random mode, and structural stochastic mode. For the cycle mode with N species, we set $a_{i,i-1} \neq 0$, resulting in a total of N physical links, as illustrated in Fig. 3(a). In the random mode with N species, we randomly selected $2N$ values of $a_{i,j}$ to be nonzero, where $i \neq j$, resulting in $2N$ random physical links, as shown in Fig. 3(c). For the structural stochastic mode, we set $a_{i,i-1} \neq 0$ for $i = 1, 2, \ldots, N$ and randomly selected an additional N values of $a_{i,j}$ to be non-zero, giving a total of $2N$ physical links [Fig. 3(e)]. For each mode with N species, we conducted 40 independent tests to assess performance in our experiments.

2. Basic tests in three-species systems

We studied a three-species system shown in Fig. 12(a) that there is a unidirectional physical link between each pair of variables. By using ECCM, three true physical links were successfully identified [Fig. 12(b)]. However, a spurious physical link in the opposite direction of the true physical link was also detected between each pair of variables. This is due to the interaction transitivity through the third-party variable. Therefore, the true physical structure cannot be accurately identified by simply applying ECCM. However, physical links are correctly identified by applying our InVaXMap method [Fig. 12(d)].

We further tested our method in three-species systems with network structure of cascading, fan-out and fan-in, respectively [Fig. 13(a)] [60,61]. As shown in Fig. 13(b), the red circles representing the presence of physical links are clearly separated from the black crosses representing absence of physical links. Therefore the threshold used to identify physical links has a wide range of options, suggesting the robustness of InVaXMap to threshold options.

3. Comparison of six methods

We tested six classical or state-of-the-art methods on the multispecies discrete Lotka-Volterra competition model across cycle, random, and structural stochastic modes. The methods tested were GC, CGC, PGC, TE, FullCI, and PCMCI. FullCI uses partial correlation (ParCorr) for conditional independence testing, which is denoted as FullCI-ParCorr. PCMCI uses linear partial correlation and two types of nonlinear independence tests (GPDC and CMI), denoted as PCMCI-ParCorr, PCMCI-GPDC, PCMCI-CMI, respectively. GPDC is based on Gaussian process regression [49] and a distance correlation [50] test on the residuals, which is suitable for a large class of nonlinear dependencies with additive noise. CMI is a fully nonparametric test based on a k-nearest neighbor estimator of conditional mutual information that accommodates almost any type of dependency [62]. To keep equal treatment for above competing methods, we used the same

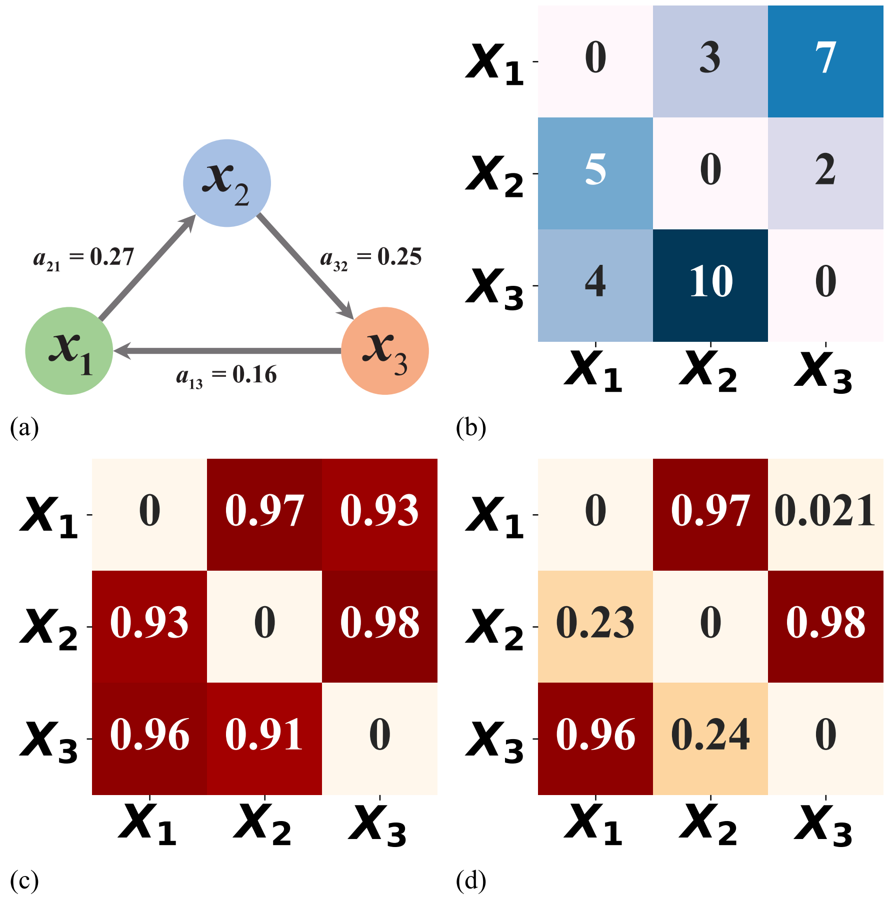


FIG. 12. A three-species system. (a) True physical structure of this three-species system. (b) Time delay matrix detected by ECCM. (c) Physical effects detected by ECCM. (d) Physical effects detected by InVaXMap using the standard partial correlation coefficient. In this system, spurious physical links are erroneously identified as direct physical relationships by the ECCM method, whereas the InVaXMap method effectively eliminates these false connections.

maximum time delay for GC, CGC, PGC, FullCI, and PCMCI as we did for InVaXMap. We carried out kNN algorithm for TE which avoided the kernel density estimation. Finally, AUC was chosen as the criterion to reduce the impact of threshold selection. The results are presented in Fig. 14 and the average AUC in three modes are recorded in Tables II, III, and IV.

The average AUC of the GC method remained around 0.5, regardless of the system size. The CGC method demonstrated some link discovery capability when the system size was 10, with an average AUC exceeding 0.6. However, as the system size increased, the average AUC dropped to approximately 0.5. The PGC method showed similar performance to CGC but failed to produce results for a system size of 100. These autoregressive-based methods mainly focus on separable systems, making them less effective for non-separable systems, such as the benchmark systems studied here.

The TE method performed well in models with cycle mode but failed in random and structural stochastic modes. FullCI and PCMCI, both developed for stochastic systems, failed to achieve an acceptable average AUC across the three modes, including the variant of PCMCI employing nonlinear independence tests.

4. Robustness test for InVaXMap

For each independent realization, the interaction strength, a_{ij}, is uniformly chosen between 0.2 and 0.4 in multi-species discrete Lotka-Volterra competition model, the interaction strength m_{ij} followed truncated normal distribution $N_T(\mu = 0, \sigma = 2I, \zeta = 0.5)$ in multispecies Ricker model, and coupling strengths α_{i-1} in synthetic gene circuit are uniformly chosen between 0.8 and 1.0. Collectively, the results across these diverse systems (Figs. 3, 20, and 30) demonstrate the robustness of InVaXMap to different coupling strengths.

To test the robustness of InVaXMap to data noise, we fixed the length of the time series to be 500 and added two types of random noise, including (i) observation noise and (ii) process noise, with different strengths in multispecies discrete Lotka-Volterra competition model. For observation

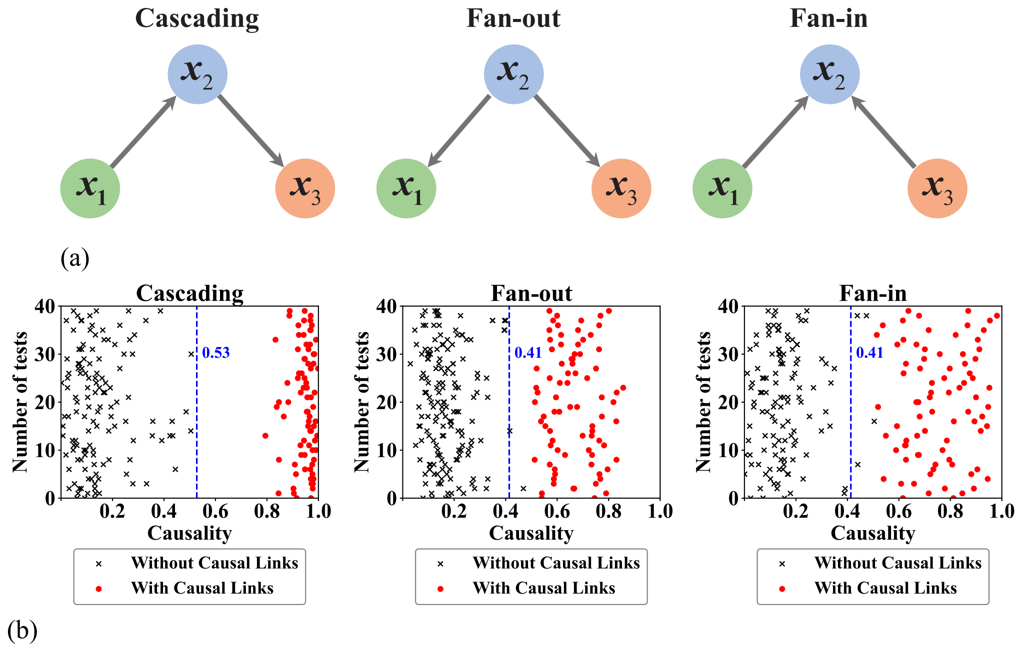


FIG. 13. (a) Three-species systems with network structure of cascading, fan-out and fan-in. (b) Physical cause-effect relations detected by InVaXMap using the standard partial correlation coefficient. The threshold used to distinguish pairs of species with causality from those without has a wide range of values, indicating the robustness of InVaXMap to different threshold settings. The black lines are the thresholds given by k-means classifier.

noises, we added noises $\hat{x}_i(t) = x_i(t) + E_{\mathrm{ob}}\varepsilon_i^{\mathrm{ob}}(t)$ to each species at each time step t, where $\varepsilon_i^{\mathrm{ob}}(t)$ subjects to a specific distribution and E_{ob} is the strength of observation noise, after one independent realization. For process noises, we modeled the intrinsic growth rate of each species i, r_i in Eq. (10), as a function of time-dependent process noises $\varepsilon_i^{\mathrm{pr}}(t)$, which was generalized by $r_i(t) = r_i[1 + E_{\mathrm{pr}}\varepsilon_i^{\mathrm{pr}}(t)]$. Noise was added with system development. For each type of noise, we produced $\varepsilon_i^{\mathrm{ob}}(t)$ and $\varepsilon_i^{\mathrm{pr}}(t)$ as two distributions, (i) Gaussian distribution $\varepsilon_i^{\mathrm{ob}}(t), \varepsilon_i^{\mathrm{pr}}(t) \sim N(0,1)$ and (ii) Laplace distribution $\varepsilon_i^{\mathrm{ob}}(t), \varepsilon_i^{\mathrm{pr}}(t) \sim La(0,1)$. We randomly selected the parameters of benchmark systems within the corresponding intervals for one independent realization, and each noise situation is conducted 40 independent realizations. The results (Fig. 15) show that InVaXMap exhibits robustness against the above two noise with different distributions across different realizations of the same model. Additonally, two types of observation noise are added to multi-species Ricker model and synthetic gene circuit. The performance of InVaXMap are well in low noise situations and declined with the noise raising (Fig. 16). InVaXMap maintains robustness under conditions of low to moderate noise, provided the underlying attractor structure remains sufficiently preserved in the reconstructed shadow manifolds. This robustness stems from the fact that cross-mapping technique relies on the continuity and smoothness of the manifold, which are not severely degraded by low-amplitude noise. The effective stability of the observed dynamics decreases (e.g., near bifurcation points or in heavily noise-driven systems), performance can degrade. The local manifold geometry in these situations is not reliable, leading to false neighbors and breaking the assumption of Takens' theorem, which in turn can cause spurious results or failures in link detection.

5. Link detection in systems with strong coupling

In addition to nonseparable systems with weak or moderate coupling, InVaXMap is also effective in systems with strong coupling. In the cycle, random, and structural stochastic modes, the interaction strength a_{ij} was randomly set between 0.8 and 0.9 for nonzero values, while all other conditions remained unchanged. As demonstrated in Fig. 17, InVaXMap exhibits strong link discovery capabilities, outperforming ECCM in these scenarios.

6. Link detection with two different forms of InVaXMap

We compared the results of link detection using two different forms of InVaXMap: the original form, defined as $\rho_{X\to Y}^{I_a} = |\mathrm{pcor}(X, X^Y | \{X^Z | Z \in I_a\})|$, and the first-order form, defined as $\rho_{1,X\to Y}^{I_a} = |\mathrm{pcor}(X, X^Y | \{X^{Z^Y} | Z \in I_a\})|$.

The information of X contained in X^{Z^Y} is also present in X^Z, with the remaining information in X^Z being orthogonal to X^Y. Therefore, replacing X^{Z^Y} with X^Z does not affect the results. To investigate this, we compared the performance of these two forms of InVaXMap in cycle mode systems with system sizes of 10, 20, and 30.

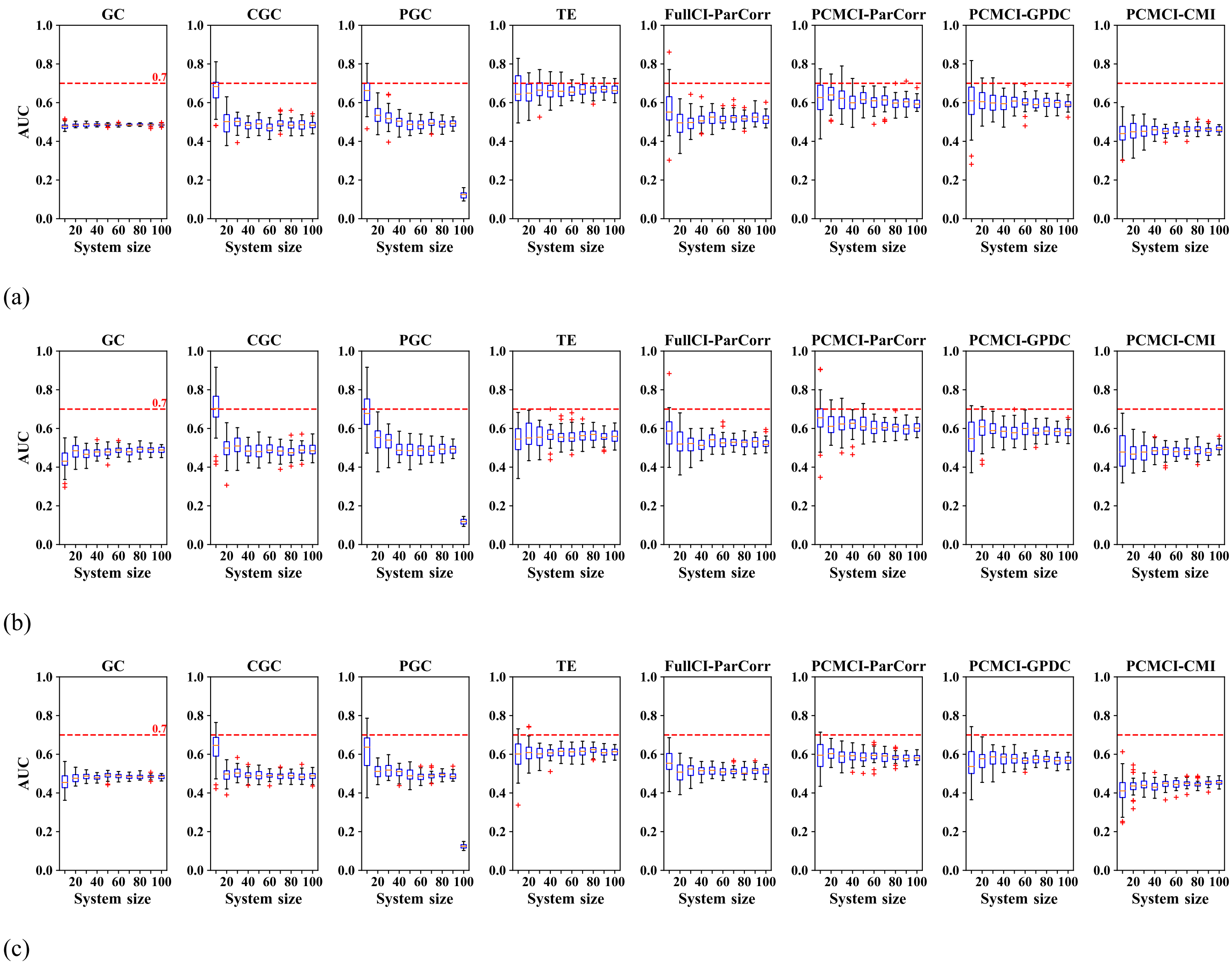


FIG. 14. Supporting AUC comparisons. Comparison results for causality detection in benchmark systems using GC, CGC, PGC, TE, FullCI(-ParCorr), and PCMCI(-ParCorr, -GPDC, -CMI). Panels (a), (b), and (c) show the link detection performance of these methods for different system sizes in cycle mode, random mode, and structural stochastic mode, respectively. None of these methods achieved a higher average AUC than InVaXMap across these three modes. The red dashed line at 0.7 serves as a visual reference threshold, though AUC performance varies across systems and configurations.

We conducted 20 independent tests for each system size, calculating the AUC for each test. As shown in Fig. 18, no significant differences were observed between the two forms of InVaXMap. The average AUC for the original form in 10-, 20-, and 30-dimensional systems were 0.740, 0.739, and 0.727, respectively. Meanwhile, the average AUC for the first-order form were 0.693, 0.717, and 0.704, respectively.

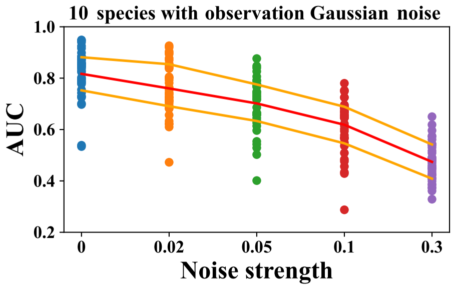
7. Additional criterion for selecting conditioning set

In the main text, we carried out incremental variable construction with respect to time delay. While the conditioning set for detecting direct physical influence is nonunique, we illustrate here an alternative criterion, based on estimation capability, for incremental variable construction.

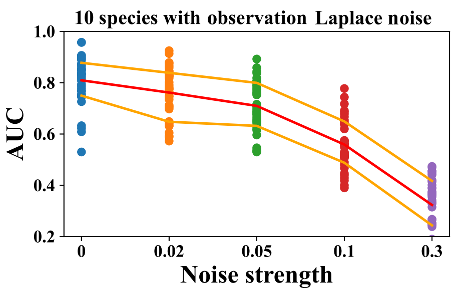
When performing the iterative cross-map-based testing, we rank variables in I_{inactive} in descending order of their estimation capability for the target variable, measured by the correlation index $\rho_{X_1\to Y}$ computed using ECCM. The conditioning set, denoted as $\tilde{I}_a$, is initialized as empty. At each iteration, we select the variable X_j with the highest reconstruction extent:

$$j = \arg\max_k \{\rho_{X_1\to X_k} | X_k \in I_{\text{inactive}}\}. \tag{E2}$$

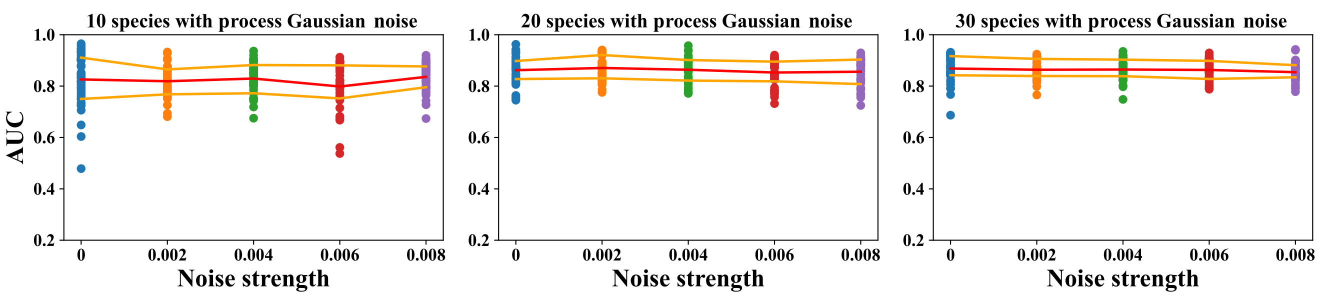
We compute the partial correlation index $\rho_{X_1\to X_j}^{\tilde{I}_a}$ to assess the relationship between X_1 and X_j while conditioning on $\tilde{I}_a$. If $\rho_{X_1\to X_j}^{\tilde{I}_a}$ exceeds a predefined threshold ϵ, X_j is added to $\tilde{I}_a$. This process is repeated until all relevant variables are identified and I_{inactive} is empty.



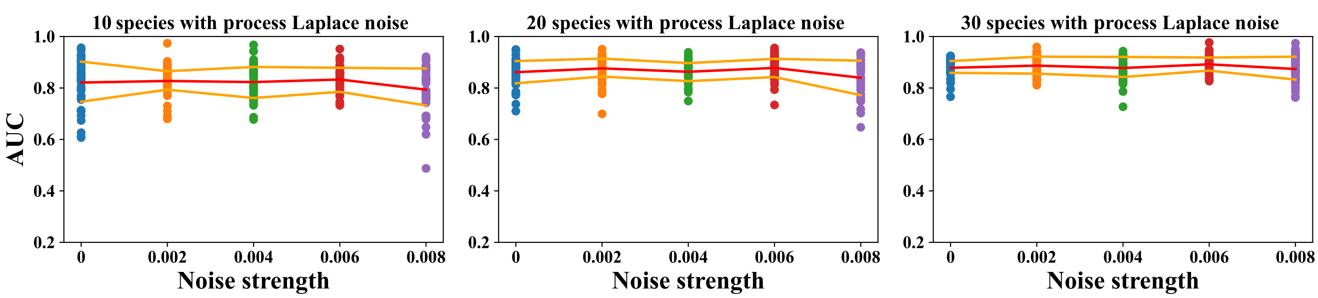
(a)



(b)



(c)



(d)

FIG. 15. Robustness of InVaXMap against random noises in multispecies discrete Lotka-Volterra competition model. The link detection ability of InVaXMap versus noise strength with observation Gaussian noise (a), observation Laplace noise (b), process Gaussian noise (c) and process Laplace noise, respectively (d). Our results show that InVaXMap exhibits robustness against the above two noise with different distributions across different realizations of the same model. It is less affected by the process noise, but more affected by the observation noise. Each result is conducted with 40 independent realizations. The red lines represent the mean of AUC and the yellow lines represent the 20% and 80% quantiles of AUC.

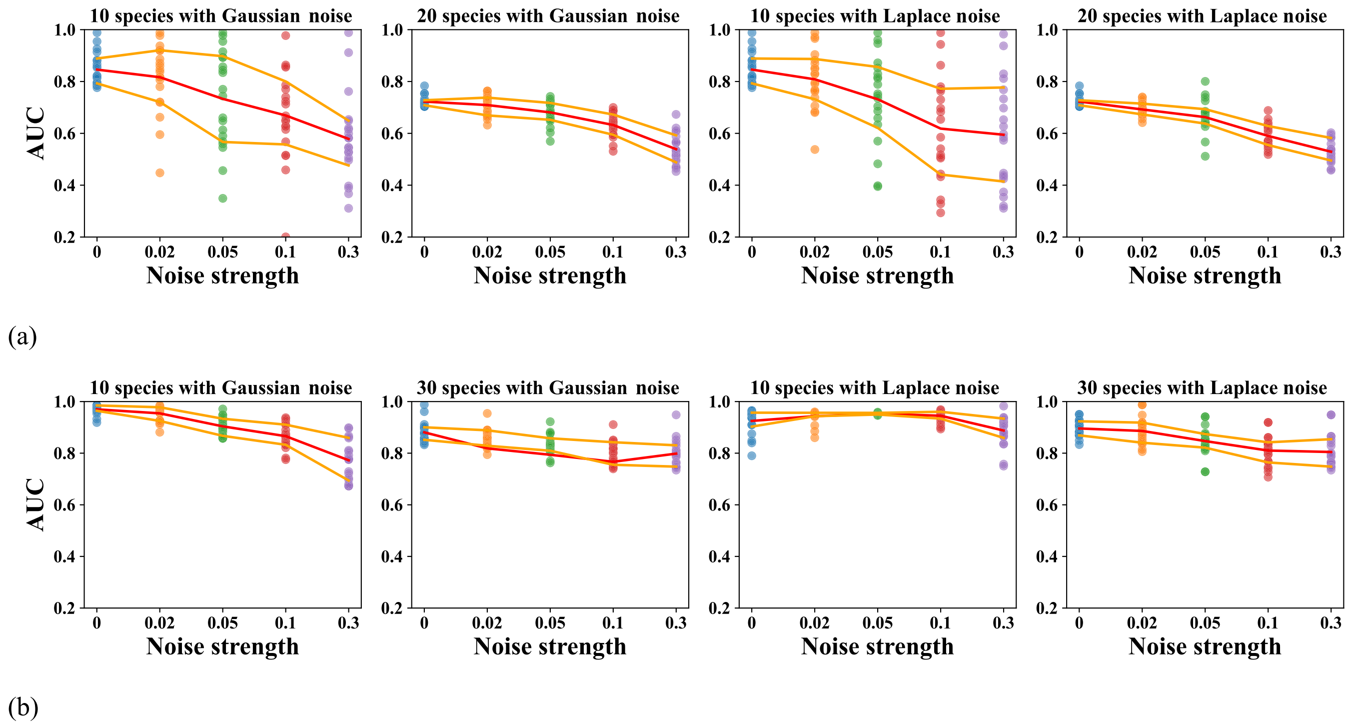


FIG. 16. Robustness of InVaXMap against observation noises. The link detection ability of InVaXMap versus noise strength with two types of observation noise in the (a) multispecies Ricker model and (b) synthetic gene circuit. The performance of InVaXMap are well in low noise situations and declined with the noise raising.

The next step, link inference via estimation capability, was then carried out with I_a replaced by $\tilde{I}_a$. We tested this criterion in the random mode of the multi-species discrete Lotka-Volterra competition model (Fig. 19). This additional criterion proved feasible within these systems, suggesting that alternative criteria for selecting conditioning sets can be explored to address specific research objectives.

8. Experiments for additional benchmark systems

We further validated the InVaXMap method through the analysis of time series data derived from two theoretical network models, multispecies Ricker model and coupled Lorenz systems, in which the interaction strengths and network are known *a priori*.

The multispecies Ricker model with n species is described as

$$N_i(t+1) = N_i(t)\exp\left[r_{0i}(1 + e_i\mathbf{M}\mathbf{N}(t))\right], \quad i = 1, \ldots, n, \tag{E3}$$

where r_{0i} is the intrinsic growth rate of species i, e_i is a vector taking zero values for all except for the ith entity, $\mathbf{M} \in R^{n\times n}$ represents interaction matrix, and $\mathbf{N}(t) = [N_1(t), \ldots, N_n(t)]^T$.

Intrinsic growth rate of species i was generated as $r_{0i} = r_{00}(1 + v_{r_0}U_1)$, where $r_{00} = 1.5$, $v_{r_0} = 0.5$ and U_1 was a random variable following uniform distribution $U(-1, 1)$. The element m_{ij} in interaction matrix $\mathbf{M}$, denoting the effect of species j on species i, was generated as

$$m_{ii} = -1, 1 \le i \le n,$$
$$m_{ij} \sim N_T(0, \sigma_I, I_{\max}) \quad 1 \le i \le n-1, i+1 \le j \le n,$$
$$m_{ij} = \begin{cases} m_{ji}(1 + v_m U_1) & \text{if } m_{ji} < 0 \\ -m_{ji}(1 + v_m U_1) & \text{else } 2 \le i \le n, 1 \le j \le i-1, \end{cases}$$

where N_T is a random variable following truncated normal distribution $N_T(\mu = 0, \sigma = \sigma I, \zeta = I_{\max})$ with probability density function involving the truncation threshold ζ,

$$f_{N_T}(N_T|\mu,\sigma,\zeta) = \frac{g(N_T)}{F_N(\zeta) - F_N(-\zeta)}, \begin{cases} g(N_T) = f_N(N_T;\mu,\sigma) & \text{if } -\zeta \le N_T \le \zeta g \\ (N_T) = 0 & \text{else,} \end{cases} \tag{E4}$$

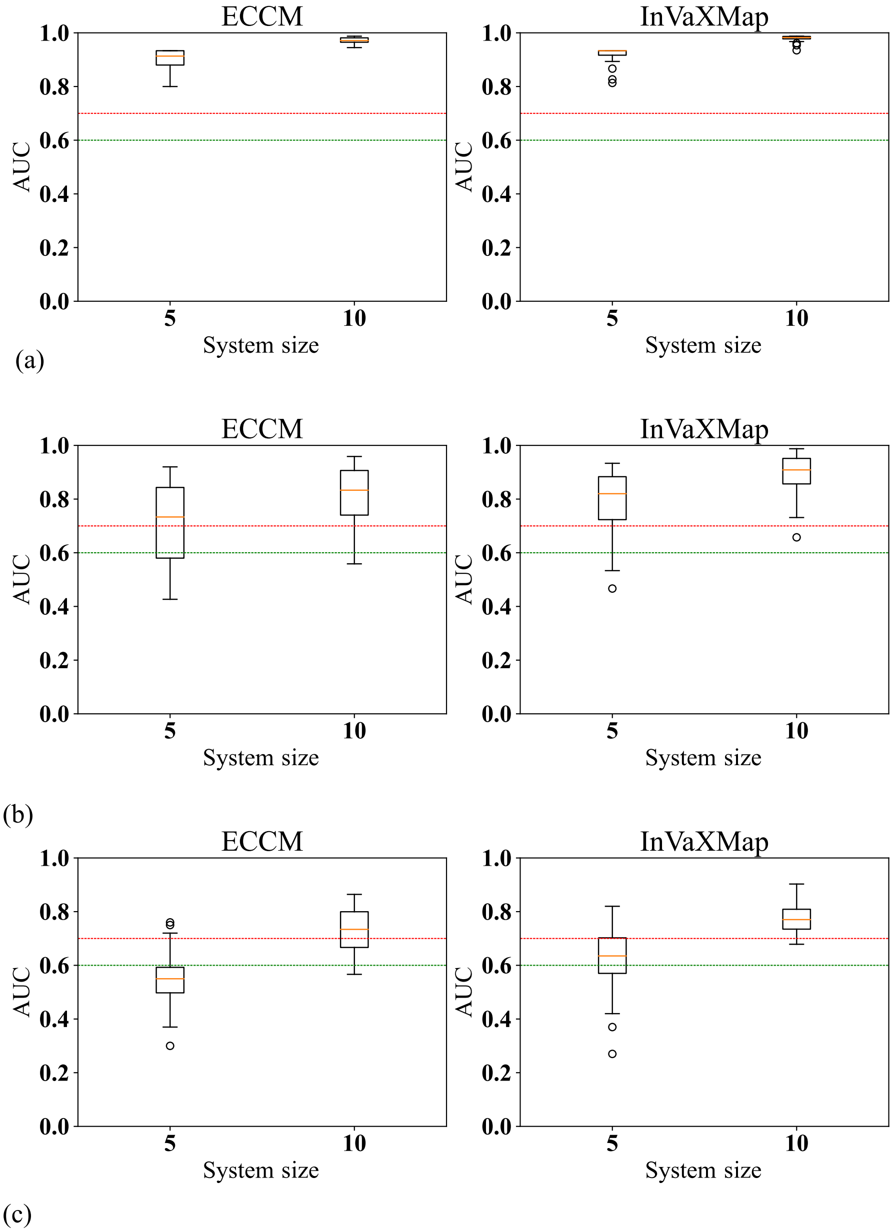


FIG. 17. Link detection in systems with strong coupling. Panels (a),(b), and (c) show the link detection performance of the four methods across different system sizes in the cycle mode, random mode, and structural stochastic mode, respectively. InVaXMap is highly effective in systems with strong coupling and outperforms other methods in link detection. The horizontal red and green dashed lines represent thresholds of 0.7 and 0.6, respectively.

where f_N and F_N are the probability density function and cumulative distribution function of normal distribution $N(\mu, \sigma)$, respectively.

Here, we let the interspecific variation of interaction strength $\sigma_I = 2$, the maximum interaction strength $I_{\max} = 0.5$, and $v_m = 0.9$ determines the maximum difference of the interaction strength between m_{ij} and m_{ji}. Finally, we generated initial population size as $N_i(t_0) = n_0(1 + N_T)$, where $n_0 = 0.1$ and N_T is a truncated normal random variable with $\mu = 0$, $\sigma = 0.2$, and $\zeta = 0.9$. The network model initially consists of 1000 interacting species. Then we randomly selected 10 (or 20) species from dominant species with mean relative abundance more than 1% (calculated over the last 100 steps). We conducted 20 independent tests for each size system [Fig. 20(a)]. In the scenario with 10 species, InVaXMap achieves the highest AUC value, 0.85, and in the scenario with 20 species, its AUC value remains relatively high, approaching 0.72. This indicates that the InVaXMap method maintains high predictive accuracy when dealing

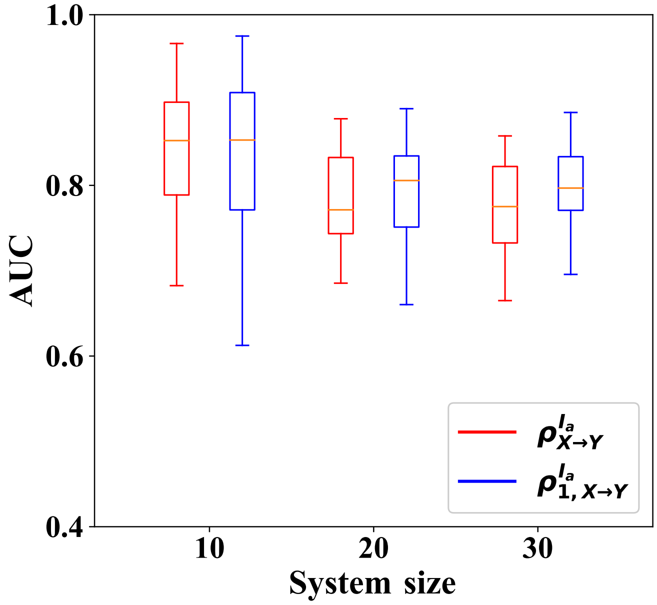


FIG. 18. Comparison of link detection methods using two variants of InVaXMap: the original form $\rho^{I_a}_{X\to Y}$ and the first-order form $\rho^{I_a}_{1,X\to Y}$. No significant differences were observed between the results of the two variants.

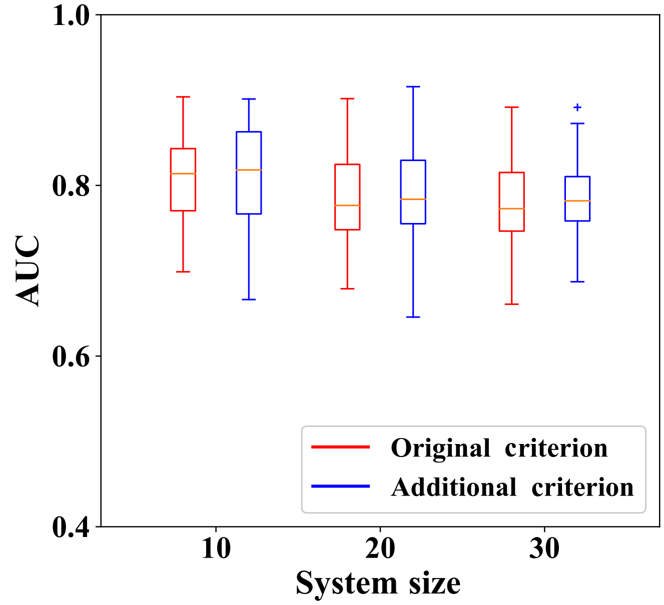


FIG. 19. AUC results of causality detection using two criteria for selecting the conditioning set. The original criterion selects variables based on time delay, while the additional criterion selects based on estimation capability. This additional criterion proved feasible within these systems, but of no significant advantage.

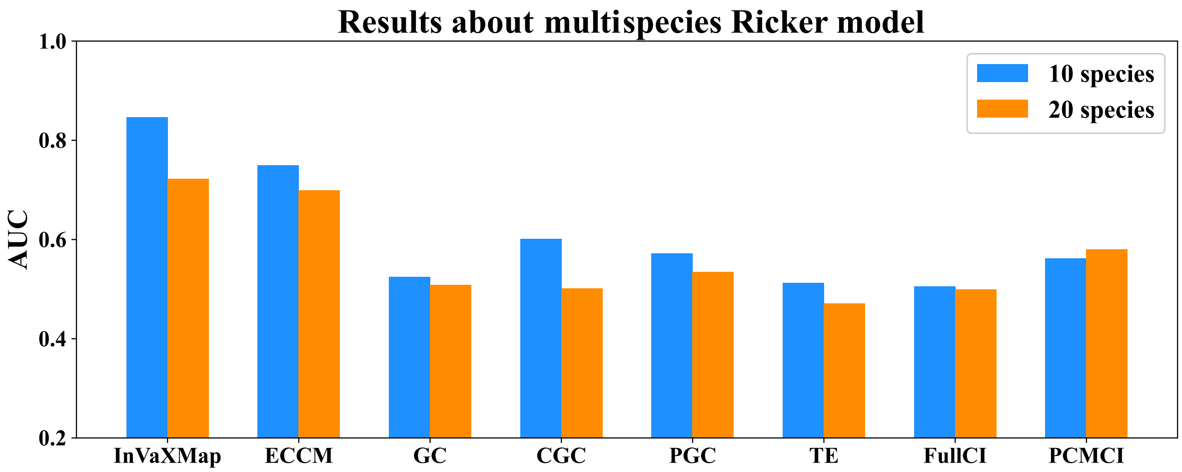
with the multispecies Ricker model. Other methods such as ECCM, GC, CGC, PGC, TE, FullCI, and PCMCI exhibit lower AUC values compared to InVaXMap in both scenarios.

The n coupled Lorenz systems [30] can be described as

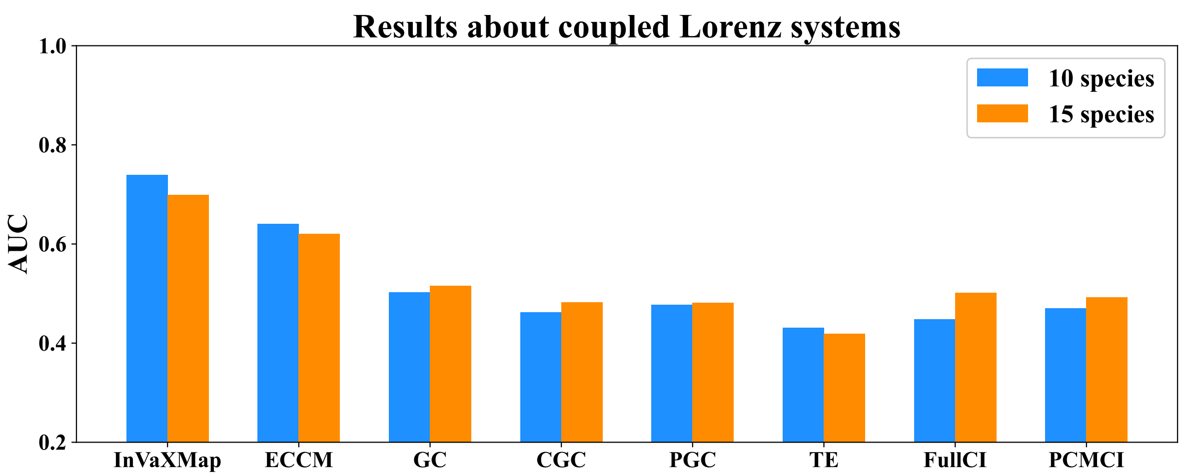
$$\dot{x}_k = \sigma(x_k - y_k) + \varepsilon^t_{k,x},$$
$$\dot{y}_k = \rho x_k - y_k - x_k z_k + \sum_{l=1}^{n} a_{kl}(x_l - y_l) + \varepsilon^t_{k,y},$$
$$\dot{z}_k = \beta z_k + x_k y_k + \varepsilon^t_{k,z}, \tag{E5}$$

where $\sigma = -10$, $\rho = 28$, $\beta = -8/3$, a_{kl} depends on the existence of a connection from the lth system to the kth system which is uniformly chosen between 0 and 0.2 if it exists, independent noise terms $\varepsilon^t_{k,d}(d = x, y, z)$ are subject to normal distribution with mean 0 and standard deviation 0.1.

Notice that the causal effect between these coupled Lorenz systems is dominated by y_k. We inferred physical relationship of these systems by the relationship between y_k. Then we set each Lorenz system to be randomly affected by one other Lorenz system and conducted 20 independent tests for $n = 10$ and 15 [Fig. 20(b)]. In those scenarios, InVaXMap achieves the highest AUC value compared to other methods such as ECCM, GC, CGC, PGC, TE, FullCI, and PCMCI.



(a)



(b)

FIG. 20. Comparison of performance for additional benchmark systems. Comparison for multispecies Ricker model and coupled Lorenz systems, across various methods. The InVaXMap method outperforms other methods in scenarios with different system sizes.

APPENDIX F: REAL-WORLD SYSTEMS IN THE MAIN TEXT

1. Brain systems

This dataset is part of a large study examining EEG (electroencephalogram) correlates of genetic predisposition to alcoholism. It includes measurements from 64 electrodes placed on subjects' scalps, sampled at 256 Hz (3.9 ms epochs) for 1 sec. We selected 10 subjects and conducted 10 trials per subject. Each trial contains time series recordings ($L = 256$) of voltage measurements on 35 electrodes (Table VI) located in the frontal, parietal, and occipital regions. Using InVaXMap, we performed controlled experiments and compared the average physical effect matrices of alcoholic brains (Fig. 21) with those of control brains (Fig. 22).

2. Gene regulatory systems

The DREAM4 dataset contains five gene regulatory networks, each consisting of 100 genes. For each gene, there are 10 realizations of 21 time series data representing gene expression. We selected 40 interacting genes from each network (Table VII) and integrated all instances into one time series for state space reconstruction. The ground

TABLE VI. List of all electrodes in each brain region.

Brain region	Electrodes
Frontal region	FP1 FP2 FPZ AF7 AF8 AF1 AF2 AFZ F7 F8 F6 F3 F4 F1 F2 FZ F5
Parietal region	CP1 CP2 CPZ CP3 CP4 PZ P1 P2 P3 P4
Occipital region	PO1 PO2 POZ PO7 PO8 O1 O2 OZ

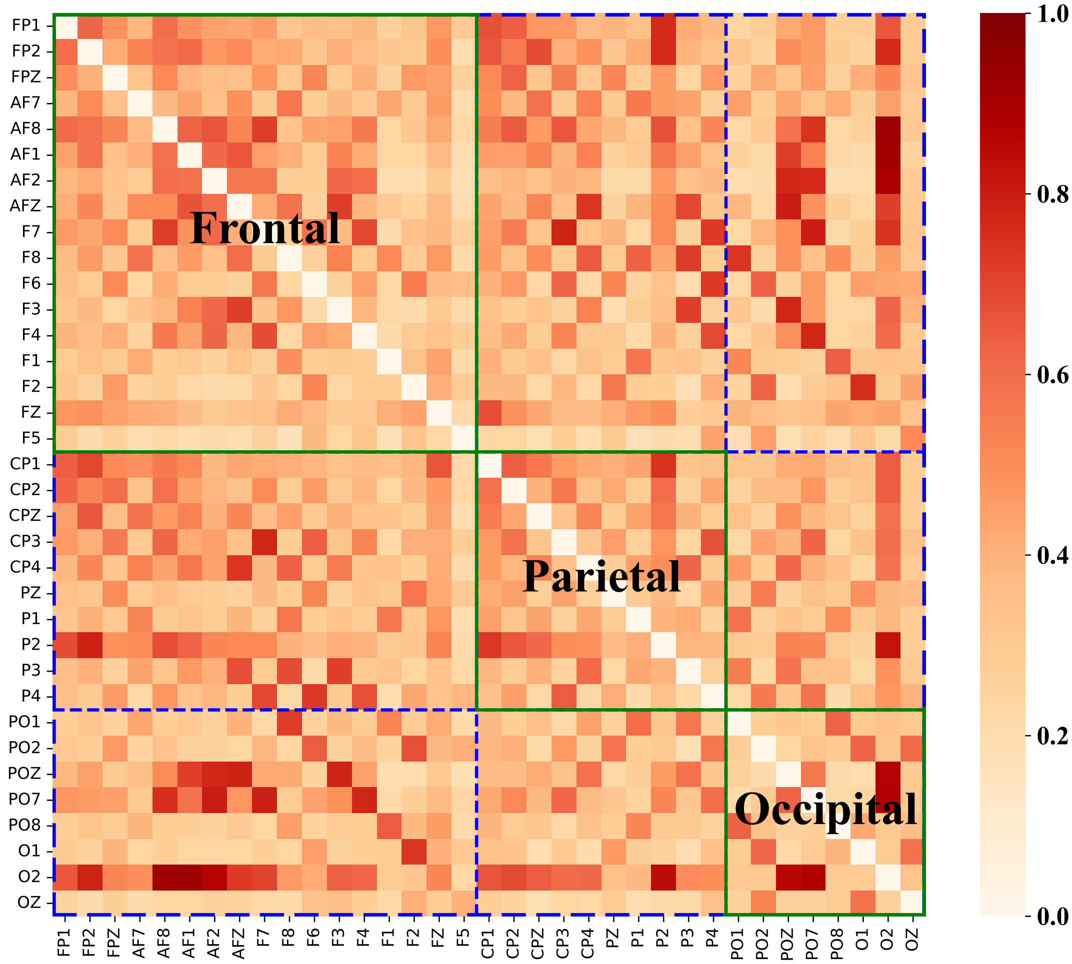


FIG. 21. Average causal effect matrix of the alcoholic brain, detected by InVaXMap using the standard partial correlation coefficient. There is a significant reduction in the strength of causal effects within and between different brain regions, which can be attributed to alcohol impairing the brain's regulatory functions both within and across regions.

truth for the selected 40 interacting genes for each network is provided in Fig. 23. Additionally, we carried out competing methods to test physical relationships in these five selected networks (Fig. 24). Compared with these method, InVaXMap reaches the highest AUC in all five selected networks.

3. Biosphere-atmosphere systems

We analyzed biosphere-atmosphere systems from four Chinese sites in the FLUXNET2015 dataset, specifically CN-Din, CN-Qia, CN-Ha2, and CN-Cha (Table VIII). Each variable in this dataset had been aggregated into daily resolutions with appropriate aggregations. Using InVaXMap, we detected links among eleven selected variables, categorized into three types: net ecosystem exchange, micrometeorology, and energy processing (Table IX). We used data from January 1st, 2003 to December 31st, 2005 ($L_{\mathrm{total}} = 1096$). First, we calculated the annual average link matrices for each of the four Chinese sites with time series length $L_{\mathrm{annual}} = 365$ or 366 (Fig. 25). Then, we computed the average link matrices for each site across the four seasons: Spring, Summer, Autumn, and Winter ($L_{\mathrm{spr}} = 92$, $L_{\mathrm{sum}} = 92$, $L_{\mathrm{aut}} = 91$ and $L_{\mathrm{win}} = 90$ or 91, Figs. 26–29).

4. Synthetic gene circuit

We additionally carried out experiments on one synthetic gene circuit, noted as the generalized repressilator model. The repressilator was designed as a simple mathematical model of transcriptional regulation. Repressor-protein concentrations, p_i, and their corresponding mRNA concentrations, m_i were treated as continuous dynamical variables. Each of these molecular species participates in transcription, translation, and degradation reactions.

$$\begin{aligned}\frac{\mathrm{d}m_i(t)}{\mathrm{d}t} &= \alpha_0 - \delta_i m_i(t) + \frac{\alpha_{i-1}K_{i-1}^n}{p_{i-1}(t)^n + K_{i-1}^n},\\ \frac{\mathrm{d}p_i(t)}{\mathrm{d}t} &= \beta_i m_i(t) - \mu_i p_i(t), \end{aligned} \tag{F1}$$

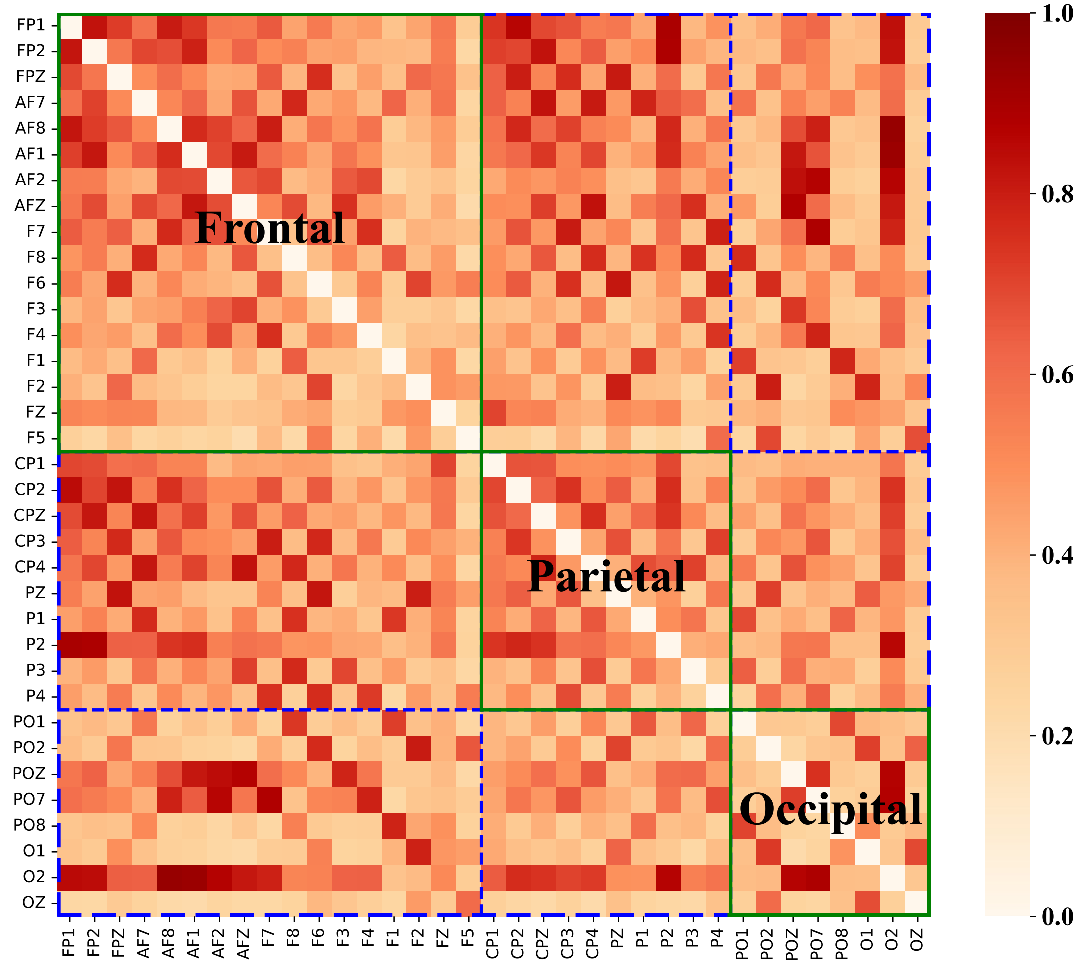


FIG. 22. Average causal effect matrix of the control brain, detected by InVaXMap using the standard partial correlation coefficient. Numerous bidirectional causal interactions between the frontal and parietal regions are observed, which are largely absent in the alcohol brain.

where $\alpha_0 = 0.009$, $\delta_i = 0.0025$, $\alpha_i \in [0.8, 1.0]$, $K_i = 40$, $n = 2$, $\beta_i = 0.05$, $\mu_i = 0.005$ and $i = 1, \ldots, N$ ($i = N$ when $i - 1 = 0$).

Once the system size was determined, we randomly generated the initial value for each gene from 0 to 1. Time series were simulated with a time step $\Delta t = 0.1$ and sampled at $t = 1$. Then the unstable sequence at the front was discarded and the following 200 points was selected as the dataset. We carried out 20 independent experiments for each system size. As shown in Fig. 30,

TABLE VII. The label of 40 selected variables from each network in DREAM4 dataset.

Network ID	Label
Network 1	4,5,10,14,15,16,19,23,25,36,37,38,43,44,45,47,50,53,54,55,56,57,61,62,67,69,72,73,75,76,77,82,83,84,85,87, 88,89,96,100
Network 2	2,4,5,7,14,18,19,21,26,28,29,30,32,33,34,36,38,46,50,53,55,59,61,63,64,65,66,70,72,77,81,82,84,89,93,94, 95,96,98,100
Network 3	2,6,9,11,12,14,15,18,19,20,22,23,24,26,32,37,38,39,43,47,48,49,52,53,54,55,57,68,70,74,79,80,81,84,89, 90,91,92,93,96
Network 4	1,3,7,11,12,16,21,25,28,31,32,34,35,37,41,42,44,46,47,49,51,54,55,58,60,61,66,67,68,69,71,75,77,78,79, 81,87,91,93,98
Network 5	1,3,4,5,7,8,9,12,13,16,17,18,19,22,23,27,34,35,36,38,39,40,41,44,50,51,53,56,57,59,61,70,72,77, 82,83,88,89,91,93

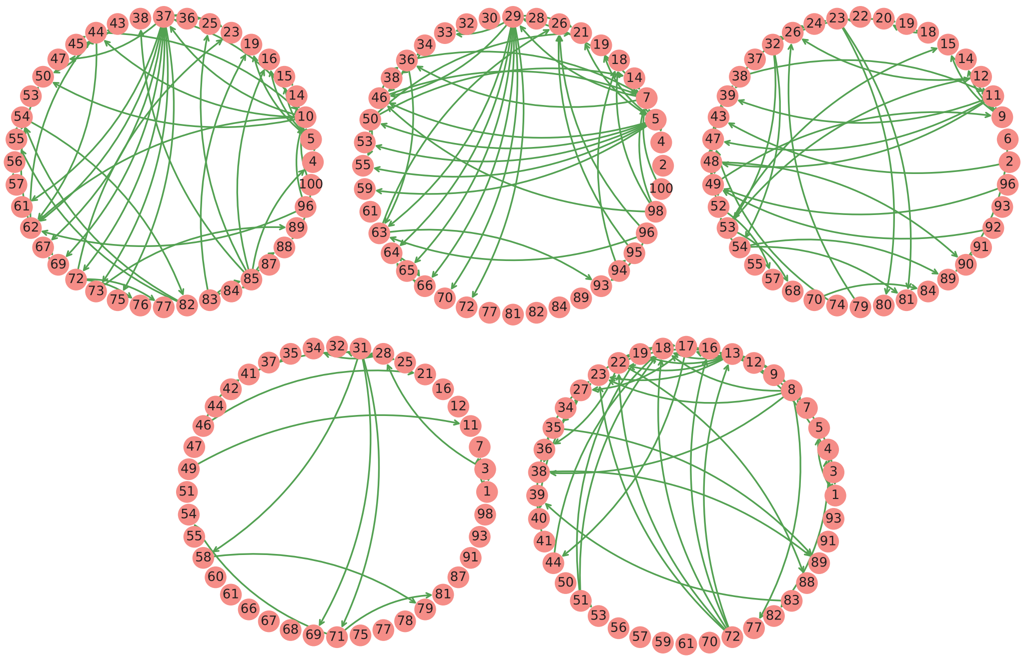


FIG. 23. Existing physical relationships of 40 selected variables from each network in DREAM4 dataset are reproduced. The success across various structures underscores the resilience of the InVaXMap method.

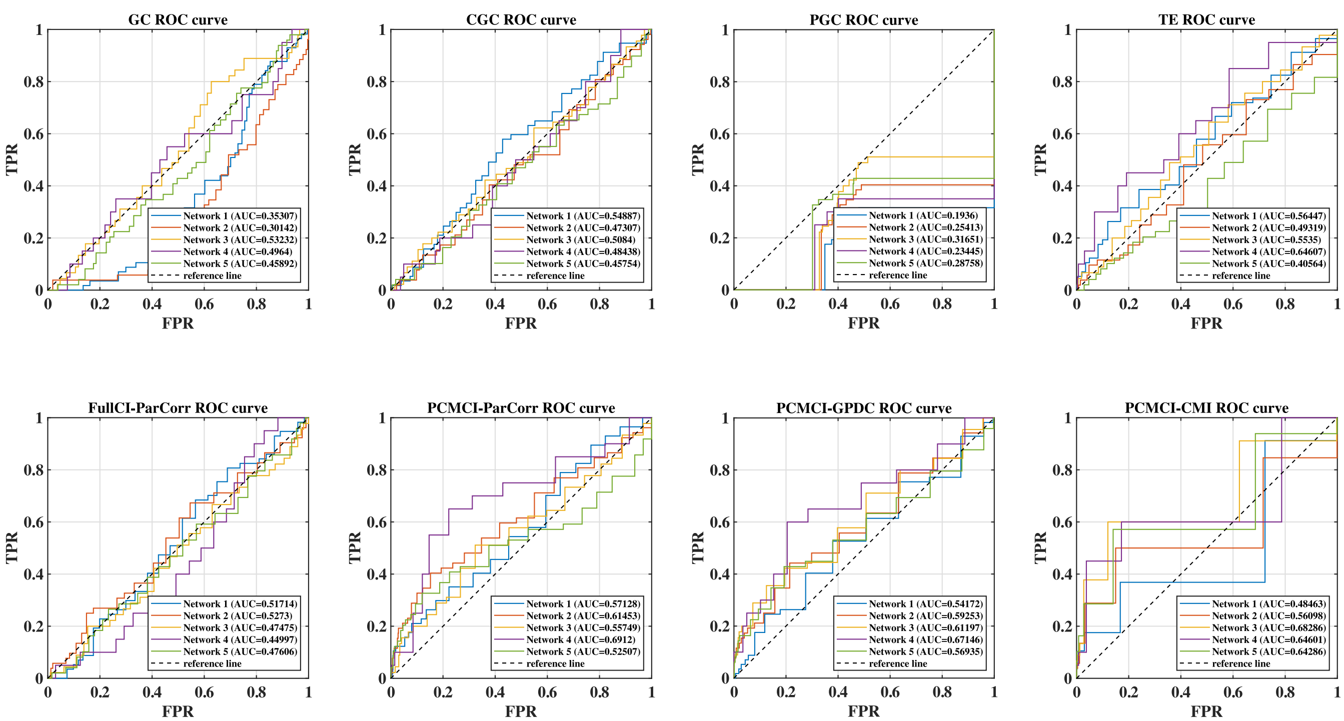


FIG. 24. Supporting AUC analysis. Results of causal detection on five gene regulatory systems using GC, CGC, PGC, TE, FullCI (-ParCorr), and PCMCI (-ParCorr, -GPDC, -CMI). None of these methods attained the average AUC across the five networks achieved by InVaXMap (0.715).

TABLE VIII. Information about four selected sites in FLUXNET2015 dataset.

SITE_ID	SITE_NAME	LOCATION_LAT	LOCATION_LONG
CN-Din	Dinghushan	23.1733	112.5361
CN-Qia	Qianyanzhou	26.7414	115.0581
CN-Ha2	Haibei Shrubland	37.6086	101.3269
CN-Cha	Changbaishan	42.4025	128.0958

TABLE IX. Information about all selected variables in FLUXNET2015 dataset.

Variable	Units	Description
	NET ECOSYSTEM EXCHANGE	
NEE	gC m-2 d-1	Net ecosystem exchange
	MICROMETEOROLOGICAL	
TA	deg C	Air temperature
SWC	%	Soil water content
VPD	hPa	Vapor pressure deficit
P	mm	Precipitation
SW	W m-2	Shortwave radiation
LW	W m-2	Longwave radiation
PA	kPa	Atmospheric pressure
WS	m s-1	Wind speed
	ENERGY PROCESSING	
LE	W m-2	Latent heat flux
H	W m-2	Sensible heat flux

the performance of InVaXMap maintained a high level in all these systems. The performance of PCM did not match that of InVaXMap in systems with 10 and 20 genes. However, when the system size reached 30, PCM outperformed InVaXMap. In these 30-dimensional systems, when PCM and InVaXMap were controlled to have the same true positive rate, InVaXMap had a higher false positive rate. This indicated that InVaXMap got weaker ability to eliminate spurious physical links compared to PCM in this situation. The performance of ECCM dropped as the system size increased, and the average AUC was around 0.54 when the system size reached 30, indicating that ECCM failed to identify real physical links.

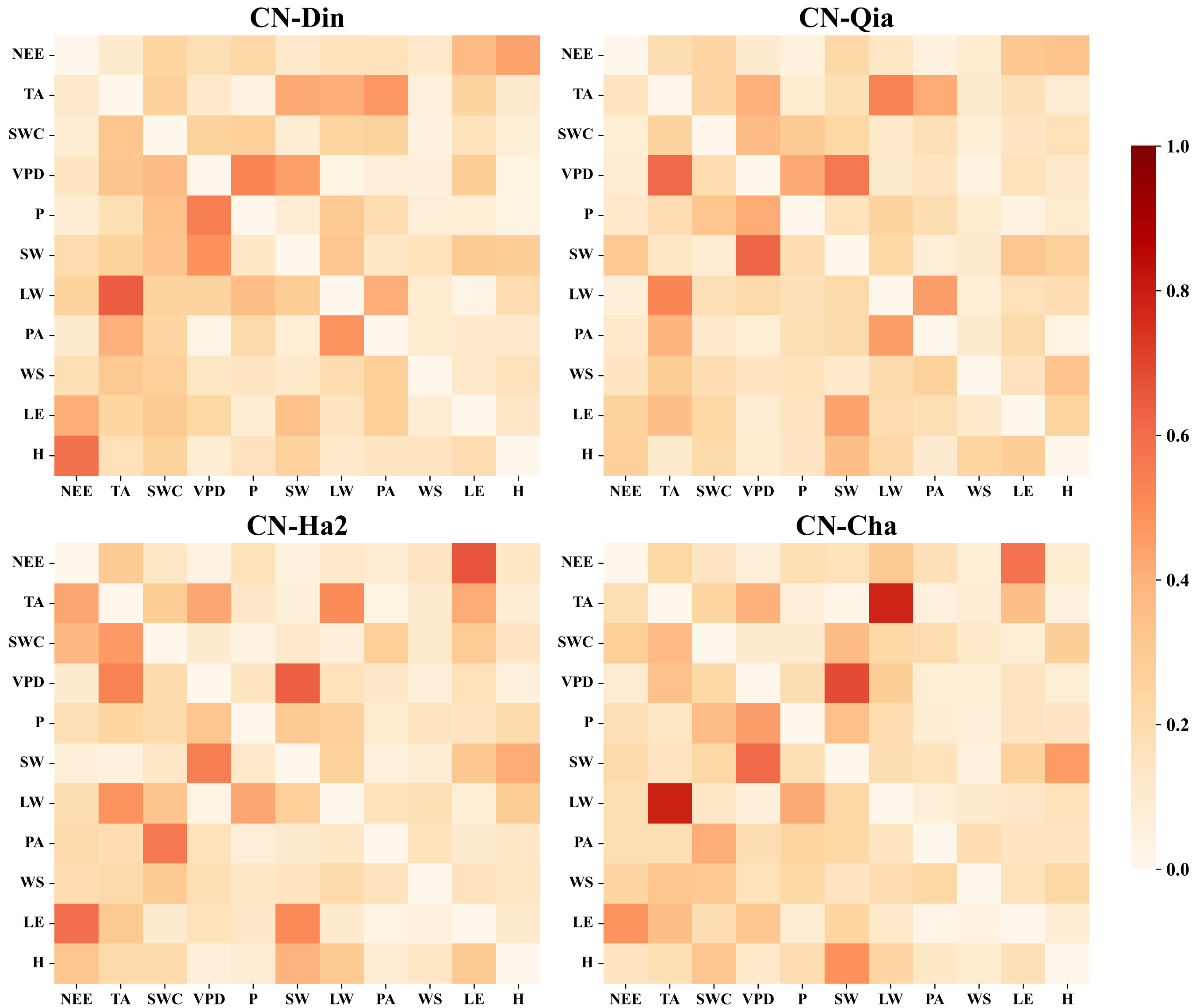


FIG. 25. Annual average link matrices for each of the four Chinese sites calculated by InVaXMap using the standard partial correlation coefficient. The geographic variability of existing physical network structures can be identified by InVaXMap.

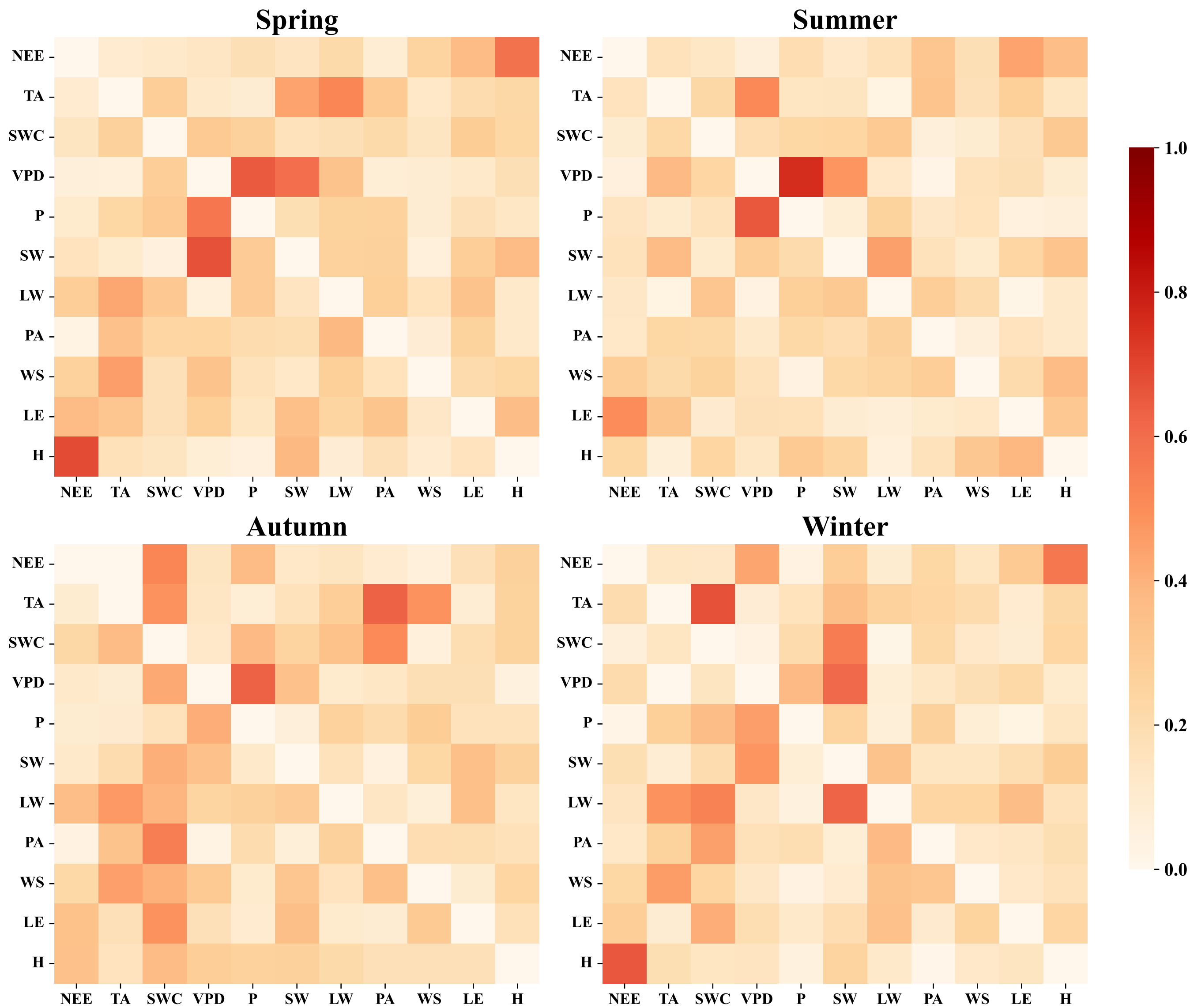


FIG. 26. Average link matrices of CN-Din site for each season calculated by InVaXMap using the standard partial correlation coefficient. The link between NEE and H is stronger in spring and winter compared to summer and autumn (in both directions). Net ecosystem exchange (NEE) is composed of ecosystem respiration (RECO) and gross primary production (GPP), which is a measure for carbon exchange between biosphere system (ecosystem) and atmosphere system. NEE is the most crucial variable in this biosphere-atmosphere system. Other variables are environmental factors. That is why, here, we focus on characteristic of causal effects between NEE and other variables. For example, we find that the physical effect between NEE and the sensible heat flux, denoted as H, is strongly affected by seasonal factors. Specifically, the link between NEE and H is stronger in spring and winter compared to summer and autumn (in both directions).

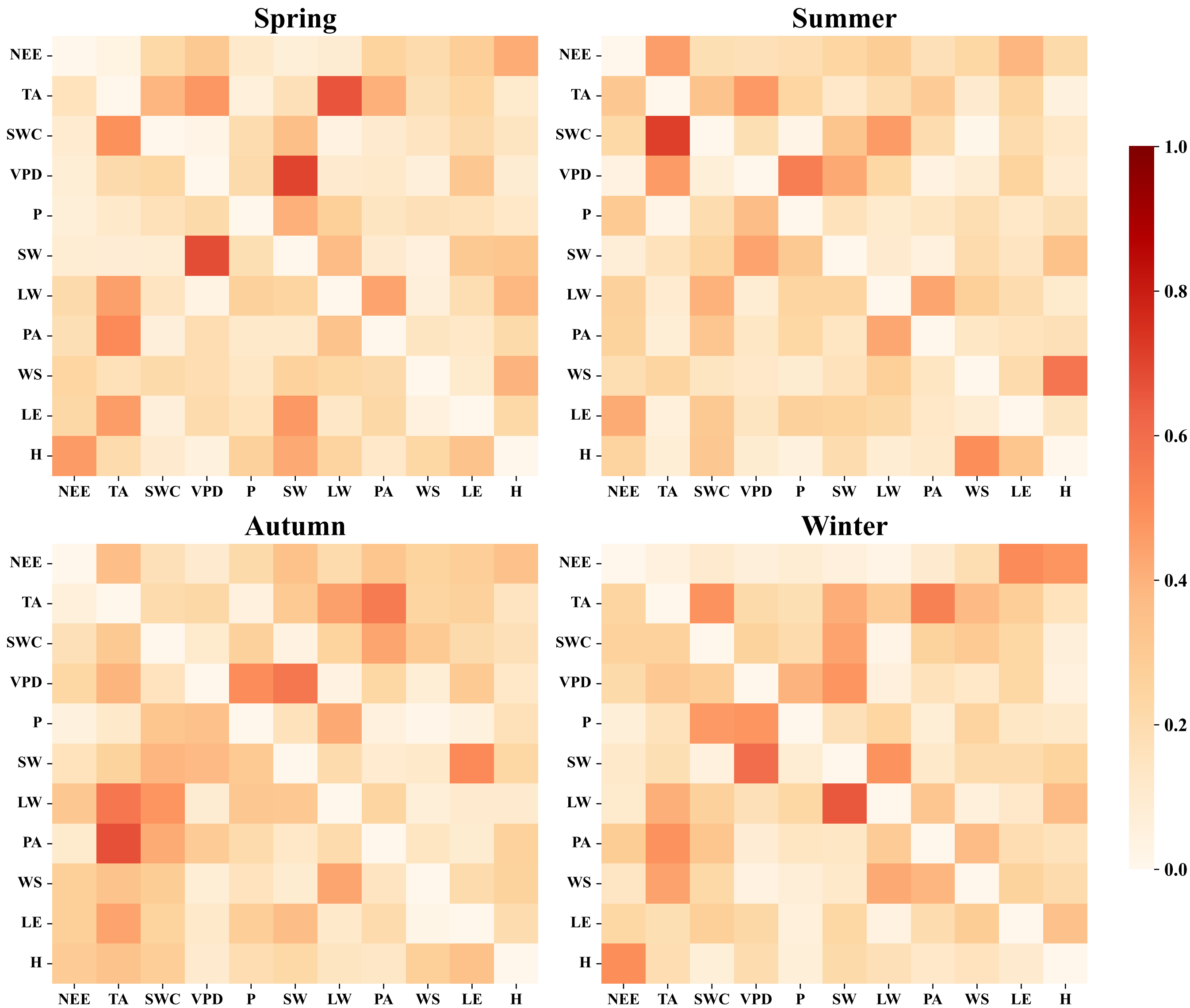


FIG. 27. Average link matrices of CN-Qia site for each season calculated by InVaXMap using the standard partial correlation coefficient. The link between NEE and H is stronger in spring and winter compared to summer and autumn.

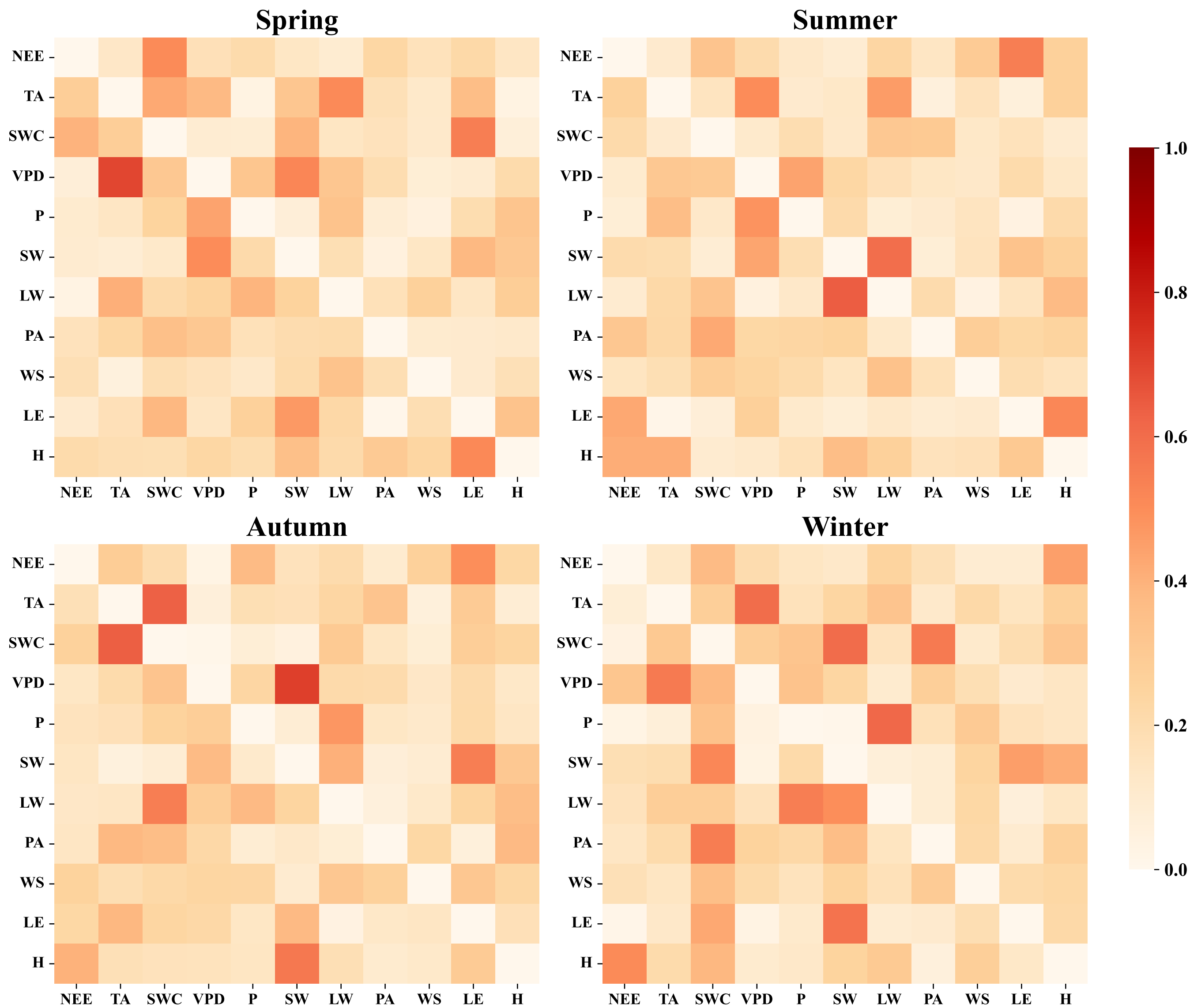


FIG. 28. Average link matrices of CN-Ha2 site for each season calculated by InVaXMap using the standard partial correlation coefficient. The link between NEE and H is stronger in summer, autumn, and winter compared to spring.

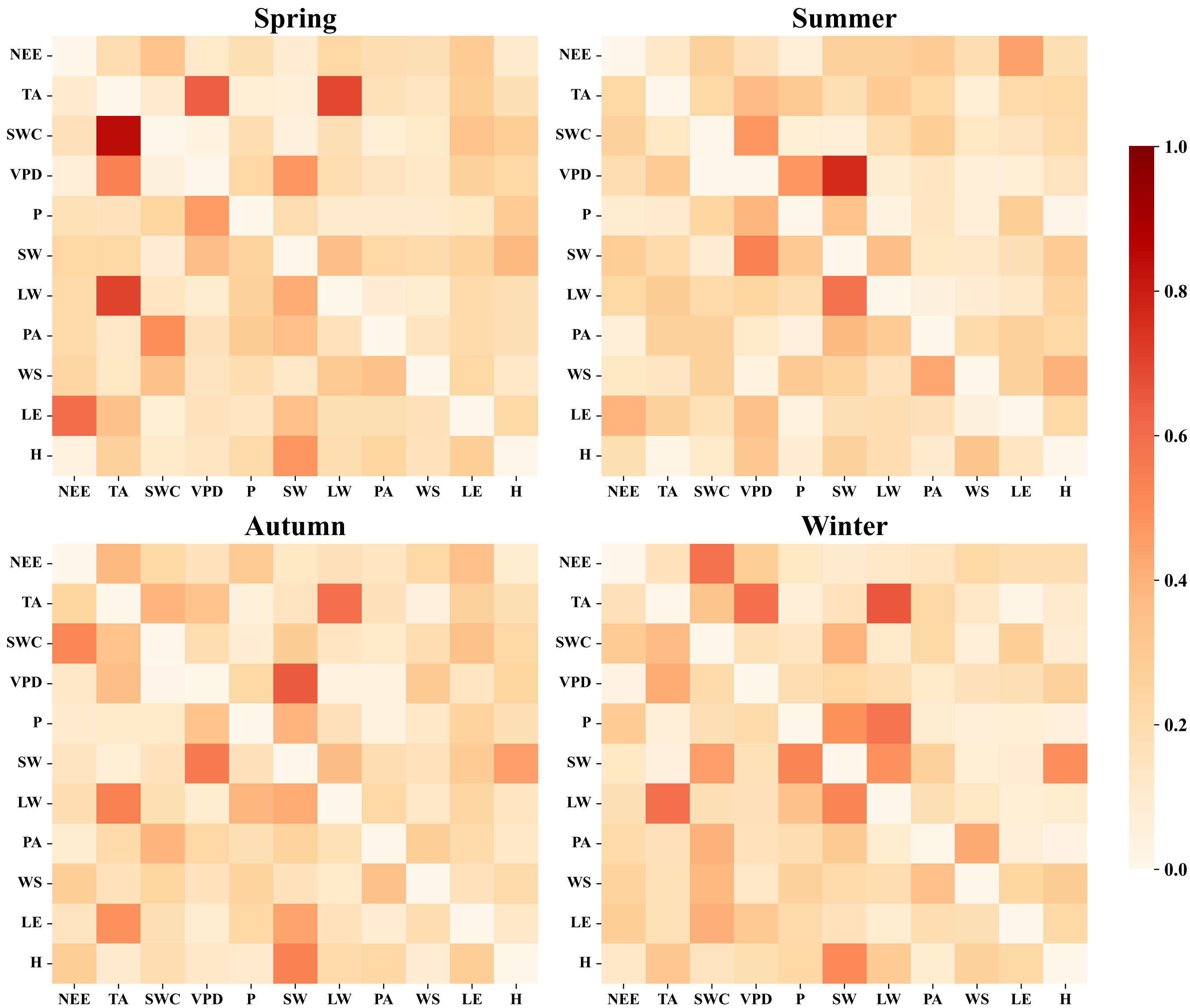


FIG. 29. Average link matrices of CN-Cha site for each season calculated by InVaXMap using the standard partial correlation coefficient. The link between NEE and H is stronger in summer and autumn compared to spring and winter. We find that the link between NEE and H varies across latitude and seasons.

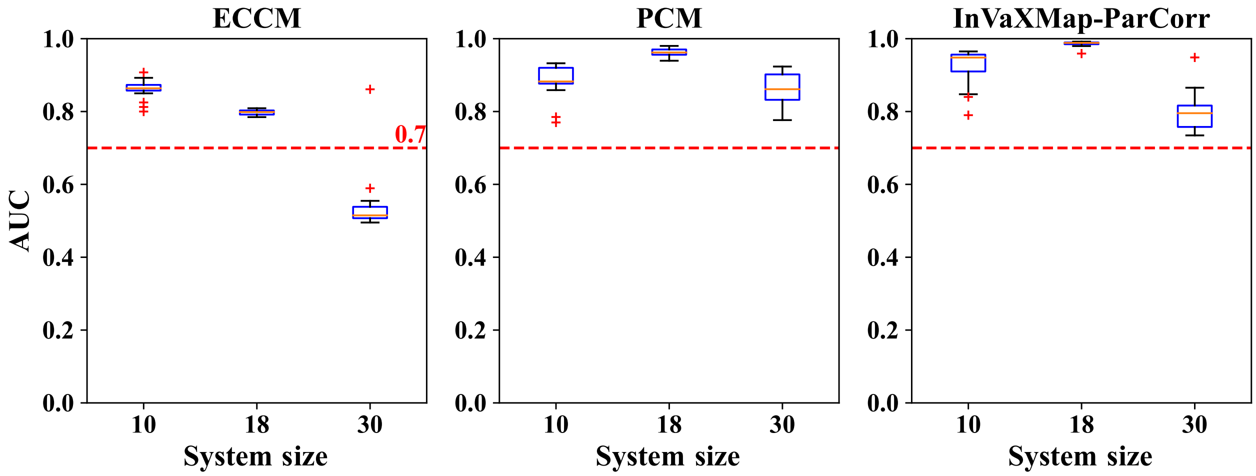
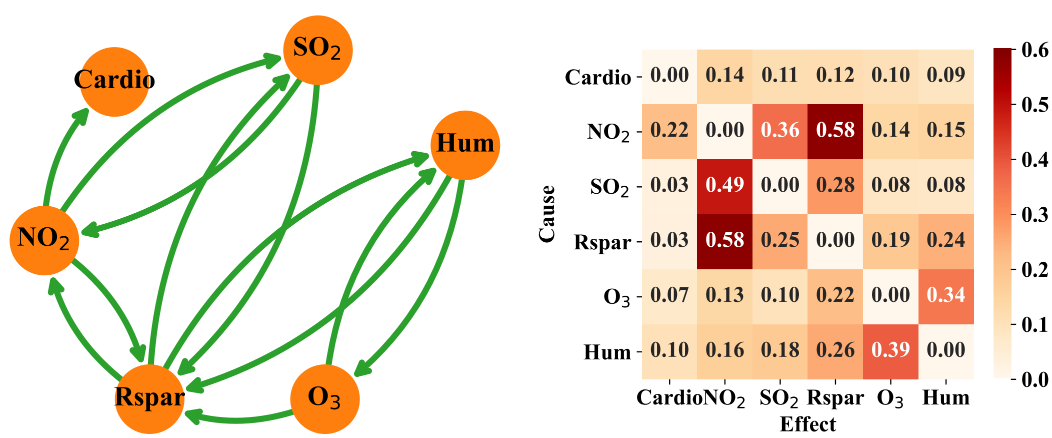


FIG. 30. Comparison of performance across ECCM, PCM, InVaXMap for synthetic gene circuit with different system sizes. The performance of InVaXMap maintained a high level in all these systems.

APPENDIX G: LINK DETECTION IN THE AIR POLLUTION DATASET USING InVaXMap

We applied InVaXMap to investigate the causal associations between daily concentrations of various air pollutants and hospitalizations for cardiovascular disease in Hong Kong from 1994 to 1997 [27]. This dataset includes daily records of admissions for cardiovascular diseases (denoted as Cardio) at major hospitals in Hong Kong. The four types of air pollutants examined are nitrogen dioxide (NO_2), sulfur dioxide (SO_2), respirable suspended particulates (Rspar), and ozone (O_3), with daily concentrations measured in $\mu g\, m^{-3}$ from monitoring stations. Additionally, we considered the physical association between relative humidity and cardiovascular disease hospitalizations.

As shown in Figs. 31(a) and 31(b), nitrogen dioxide was identified as the major cause of cardiovascular disease hospitalizations. However, no significant physical influence was detected for the other pollutants or relative humidity. Our analysis also revealed bidirectional physical relationships between several pairs of pollutants, including nitrogen dioxide, sulfur dioxide, and respirable suspended particulates, as well as a bidirectional physical relationship between ozone and humidity. These findings are consistent with previous studies [63–65].



(a) (b)

FIG. 31. Link detection in the air pollution dataset calculated by InVaXMap using the standard partial correlation coefficient. (a) Reconstructed physical network. (b) Numerical results of InVaXMap in the air pollution system. Link indices greater than 0.2 are considered as true physical relationships. The network shows that nitrogen dioxide (NO_2) is identified as the primary cause of cardiovascular disease, along with several bidirectional physical relationships.

APPENDIX H: CODE AVAILABILITY

The codes for InVaXMap that we developed in this article are publicly available at https://github.com/Atticusl/Incremental-Variable-Construction-based-Cross-Mapping.

APPENDIX I: DATA SOURCES

We used four datasets of real-world systems to validate our method. The first dataset is brain-activity data from Henri Begleiter's Neurodynamics Laboratory at the State University of New York Health Center in Brooklyn [23]. The second dataset consists of gene regulatory systems based on the DREAM4 gene expression data [24]. The third dataset is derived from the FLUXNET2015 biosphere-atmosphere datasets [25]. The fourth dataset includes air pollution and hospital admissions for cardiovascular diseases in Hong Kong, sourced from [27].

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